The p-adic upper half plane

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Abstract

In this report, we investigate the geometry of the p-adic upper half plane (the Drinfeld half plane) defined over a non-Archimedean field K.

We begin with a brief introduction to rigid spaces and formal schemes to establish the necessary background knowledge. We then present three perspectives for studying this object: the rigid analytic viewpoint, the formal scheme viewpoint, and the functorial viewpoint. Finally, we describe the action of $\mathrm{PGL}_2(K)$ on these respective viewpoints.

In the second part, we primarily follow the first part of the classic article [2], supplementing it with detailed derivations that were omitted in the original text, including explicit calculations of coordinate rings, details on the gluing of schemes, and clarifications of ambiguous statements.

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Part I
preliminary

Let $\mathbb{K}$ be a completed non-archimedean valuation field (valued in $\mathbb{R}$), and $\mathbb{O}$ is its ring of integers. Without further explanation, we always denote the norm as $|\cdot|$ and the valuation as v.

In this part, we omit most of the proof and details in order to give a brief introduction to rigid spaces and formal schemes.

1 An introduction to Rigid Geometry

1.1 Affinoid spaces (simple version)

Definition 1.1 (Tate algebras)**.** *The Tate algebra of degree n, T_n, is the algebra of convergent power series of convergence radius* 1*.*

To be more explicit,

$$T_n := \left\{ \sum_{v \in V_n} a_v T^v \;\middle|\; a_v \in \mathbb{K},\ \lim_{|v| \to +\infty} a_v = 0 \right\},$$

where $V_n = \mathbb{N}^n$, and for each $v = (v_1, \cdots, v_n) \in V_n$,

$$|v| = v_1 + \cdots + v_n, \quad T^v = T_1^{v_1} \cdots T_n^{v_n}.$$

Definition 1.2 (Affinoid algebras)**.** *A commutative ring A is an affinoid algebra if there is a number $n \in \mathbb{N}_+$ and a surjective $\mathbb{K}$- homomorphism $\alpha : T_n \twoheadrightarrow A$.*

Proposition 1.1.1. *Let $\phi : A \to B$ be a morphism between affinoid algebras. Then for any maximal ideal $\mathfrak{m} \subseteq B$, $\phi^{-1}(\mathfrak{m})$ is a maximal ideal of A.*

Definition 1.3 (Affinoid spaces associated to affinoid algebras)**.** *Let A be an affinoid algebra, then we call $\operatorname{Sp} A$ the associated affinoid space of A, where $\operatorname{Sp} A$ is the collection of all maximal ideals of A equipped with the Zariski topology, i.e.*

$$D(f) = \{\mathfrak{m} \in \operatorname{Sp} A \mid f \notin \mathfrak{m}\},\ f \in A$$

forms a basis of the topology.

By the above proposition, it is clear that a morphism $\phi : A \to B$ between affinoid algebras induces a unique map $\tilde{\phi}$ from $\operatorname{Sp} B$ to $\operatorname{Sp} A$:

$$\tilde{\phi}(\mathfrak{m}) := \phi^{-1}(\mathfrak{m}) \in \operatorname{Sp} A, \quad \forall \mathfrak{m} \in \operatorname{Sp} B$$

We temporarily define the morphisms between the affinoid spaces as the morphisms induced by their associated affinoid algebras.

Proposition 1.1.2. *For any $\mathfrak{m} \in \operatorname{Sp} A$, $A/\mathfrak{m}$ is an algebraic extension of $\mathbb{K}$.*

Then $|f(x)| \in \mathbb{R}_{\geq 0}$ is well-defined, where $f \in A$ and $x = \mathfrak{m} \in \operatorname{Sp} A$, and we view $f(x)$ as the image of f in $A/\mathfrak{m}$.

Definition 1.4 (Affinoid subdomains)**.** *Let $X = \operatorname{Sp} A$ be an affinoid space. A subset $U \subseteq X$ is called an affinoid subdomain of X if there exist an affinoid space $X' = \operatorname{Sp} A'$ and a morphism of affinoid spaces $\iota : X' \longrightarrow X$ such that $\iota(X') \subseteq U$ and the following universal property holds: Any morphism of affinoid spaces $\varphi : Y \longrightarrow X$ satisfying $\varphi(Y) \subset U$ admits a unique factorization through $\iota : X' \longrightarrow X$ via a morphism of affinoid spaces $\varphi' : Y \longrightarrow X'$.*

Write $\mathcal{O}_X(U) := A'$. Clearly, $\mathcal{O}_X(X) = A$. And we have the following results:

Theorem 1.1.3. *1. The morphism $X' \xrightarrow{\iota} U$ is a bijection.*

2. The intersection of two affinoid subdomains is still an affinoid subdomain.

Theorem 1.1.4 (Tate)**.** *Let $X = \operatorname{Sp} A$ be an affinoid space. $\{U_i\}_{i=1}^n$ is a collection of finite many affinoid subdomains of X such that*

$$X = \bigcup_{i=1}^{n} U_i.$$

Then the following sequence is exact:

$$\mathcal{O}_X(X) \longrightarrow \prod_i \mathcal{O}_X(U_i) \rightrightarrows \prod_{i,j} \mathcal{O}_X(U_i \cap U_j)$$

1.2 Rigid spaces

Definition 1.5 (completed Grothendieck topology)**.** *Let X be a non-empty set, then a completed Grothendieck topology (G-topology) on X, τ, consists of*

1. *a collection of subsets of X which we denote by $O\tau$,*
2. *a collection of pairings $(U, \{U_i\}_{i\in I})$ such that $U, U_i \in O\tau$ and $U = \bigcup_i U_i$ which we denote by $Cov\tau$.*

And τ satisfies the following conditions:

1. *$\{(U, U) \mid U \in O\tau\} \subseteq Cov\tau$;*
2. *If $(U, \{U_i\}_i), (U_i, \{V_{ij}\}_j) \in Cov\tau \quad \forall i$, then $(U, \{V_ij\}_{i,j}) \in Cov\tau$;*
3. *$\varnothing, X \in O\tau$;*
4. *If $(U, \{U_i\}_i) \in Cov\tau, V \subset U$ and $V \cap U_i \in O\tau \ \forall i$, then $V \in O\tau$;*
5. *Let $U, U_i \in O\tau \ i \in I$,and $U = \bigcup_i U_i$. If there exists a refinement $\{V_j\}_j$ of $\{U_i\}_i$ such that $(U, \{V_j\}_j) \in Cov\tau$, then $(U, \{U_i\}_i) \in Cov\tau$.*

We then call (X, τ) a G-topological space.

In a word, we introduce the Grothendieck topology to focus on the nice opens and the nice coverings.

Definition 1.6. *Let X be an affinoid space. The strong Grothendieck topology, τ_X, on X is given as follows. (It is easy to check that τ_X is a G-topology on X.)*

1. *A subset $U \subset X$ is called admissible open (i.e. contained in $O\tau_X$) if there is a covering $U = \bigcup_{i\in I} U_i$ of U by affinoid subdomains $U_i \subset X$ such that for all morphisms of affinoid spaces $\varphi : Z \to X$ satisfying $\varphi(Z) \subset U$ the covering $\{\varphi^{-1}(U_i)\}_{i\in I}$ of Z admits a refinement that is a finite covering of Z by affinoid subdomains.*
2. *A covering $V = \bigcup_{j\in J} V_j$ of some admissible open subset $V \subset X$ is called an admissible covering (i.e. $(V, \{V_j\}_j) \in Cov\tau_X$) if for each morphism of affinoid spaces $\varphi : Z \longrightarrow X$ satisfying $\varphi(Z) \subset V$ the covering $(\varphi^{-1}(V_j))_{j\in J}$ of Z admits a refinement that is a finite covering of Z by affinoid subdomains.*

Theorem 1.2.1 (structure sheaf)**.** *Let X be an affinoid space with the strong Grothendieck topology. Then there is a unique sheaf $\mathcal{F}$ on X with respect to the G-topology such that $\mathcal{F}(U) = \mathcal{O}_X(U)$ for all affinoid subdomain $U \subseteq X$. And we also denote $\mathcal{F}$ by $\mathcal{O}_X$ called the structure sheaf on X.*

And from now on, when we say an affinoid space, it actually means $(X, \tau_X, \mathcal{O}_X)$, an affinoid space with the strong Grothendieck topology and the structure sheaf.

Definition 1.7. *A G-ringed space is a triple $(X, \tau, \mathcal{O}_X)$ where (X, τ) is a G-topological space and $\mathcal{O}_X$ a sheaf of $\mathbb{K}$-algebras on it. $(X, \tau, \mathcal{O}_X)$ is called a locally G-ringed space if, in addition, all stalks $\mathcal{O}_{X,x}$, $x \in X$, are local rings.*

Definition 1.8 (Rigid spaces)**.** *A rigid space is a locally G-ringed space $(X, \tau, \mathcal{O}_X)$ such that X admits an covering $\{X_i\}_{i\in I}$ such that $(X, \{X_i\}_i) \in Cov\tau$ and $(X_i, \tau|_{X_i}, \mathcal{O}_X|_{X_i})$ is an affinoid space for all $i \in I$.*

1.3 The geometry of rigid spaces

We will introduce some geometry of rigid spaces.

Definition 1.9. *Let $X = \operatorname{Sp} A$ be an affinoid space.*

1. *A subset in X of type*
$$X(f_1, \ldots, f_r) = \{x \in X \mid |f_i(x)| \leq 1\}$$
for functions $f_1, \ldots, f_r \in A$ is called a Weierstrass domain in X.

2. *A subset in X of type*
$$X(f_1, \ldots, f_r, g_1^{-1}, \ldots, g_s^{-1}) = \{x \in X \mid |f_i(x)| \leq 1, |g_j(x)| \geq 1\}$$
for functions $f_1, \ldots, f_r, g_1, \ldots, g_s \in A$ is called a Laurent domain in X.

3. *A subset in X of type*
$$X\left(\frac{f_1}{f_0}, \ldots, \frac{f_r}{f_0}\right) = \{x \in X \mid |f_i(x)| \leq |f_0(x)|\}$$
for functions $f_0, \ldots, f_r \in A$ without common zeros is called a rational domain in X.

Proposition 1.3.1. *Weierstrass domains, laurent domains and rational domains are affinoid subdomains. In the following, we will give their corresponding affinoid algebras.*

1. *For the Weierstrass domain $X(f_1, \ldots, f_r)$, its corresponding affinoid algebra is*
$$A\langle f_1, \ldots, f_r\rangle := A\langle\zeta_1, \ldots, \zeta_r\rangle/(\zeta_i - f_i; i = 1, \ldots, r).$$

2. *For the Laurent domain $X(f_1, \ldots, f_r, g_1^{-1}, \ldots, g_s^{-1})$, its corresponding affinoid algebra is*
$$\begin{aligned} &A\langle f_1, \ldots, f_r, g_1^{-1}, \ldots, g_s^{-1}\rangle := \\ &\quad A\langle\zeta_1, \ldots, \zeta_r, \xi_1, \ldots, \xi_s\rangle/(\zeta_i - f_i, 1 - g_j\xi_j; i = 1, \ldots, r; j = 1, \ldots, s). \end{aligned}$$

3. *For the Laurent domain $X\left(\frac{f_1}{f_0}, \ldots, \frac{f_r}{f_0}\right)$, its corresponding affinoid algebra is*
$$A\left\langle \frac{f_1}{f_0}, \ldots, \frac{f_r}{f_0}\right\rangle := A\langle\zeta_1, \ldots, \zeta_r\rangle/(f_i - f_0\zeta_i; i = 1, \ldots, r)$$

Thus, they are admissible open subsets. An admissible open subset $U \subset X$ that the induced locally G-ringed space $(U, \mathcal{O}_X|_U)$ is a rigid space again; we will call $(U, \mathcal{O}_X|_U)$ an open subspace of $(X, \mathcal{O}_X)$. As usual, global rigid spaces can be constructed by gluing local ones.

Proposition 1.3.2 (gluing rigid spaces)**.** *Consider the following data:*

(i) rigid spaces $X_i, i \in I$, and

(ii) open subspaces $X_{ij} \subset X_i$ and isomorphisms $\varphi_{ij} \colon X_{ij} \xrightarrow{\sim} X_{ji}$, for all $i, j \in I$,

and assume that these are subject to the following conditions:

(a) $\varphi_{ij} \circ \varphi_{ji} = \mathrm{id}$, $X_{ii} = X_i$, and $\varphi_{ii} = \mathrm{id}$, for all $i, j \in I$,

(b) φ_{ij} induces isomorphisms $\varphi_{ijk} \colon X_{ij} \cap X_{ik} \xrightarrow{\sim} X_{ji} \cap X_{jk}$ that satisfy $\varphi_{ijk} = \varphi_{kji} \circ \varphi_{ikj}$ for all $i, j, k \in I$.

Then the X_i can be glued by identifying X_{ij} with X_{ji} via φ_{ij} to yield a rigid space X admitting $(X_i)_{i \in I}$ as an admissible covering.

More precisely, there exists a rigid space X together with an admissible covering $(X'_i)_{i \in I}$ and isomorphisms $\psi_i \colon X_i \xrightarrow{\sim} X'_i$ restricting to isomorphisms $\psi_{ij} \colon X_{ij} \xrightarrow{\sim} X'_i \cap X'_j$ such that the diagram

$$\begin{array}{ccc} X_{ij} & \xrightarrow{\psi_{ij}} & X'_i \cap X'_j \\ {\scriptstyle \varphi_{ij}}\downarrow & & \| \\ X_{ji} & \xrightarrow{\psi_{ji}} & X'_j \cap X'_i \end{array}$$

is commutative. Furthermore, X is unique up to canonical isomorphism.

More easy is the gluing of morphisms.

Proposition 1.3.3 (gluing morphisms)**.** *Let X, Y be rigid spaces and let $(X_i)_{i \in I}$ be an admissible covering of X. Furthermore, let $\varphi_i \colon X_i \longrightarrow Y$ be morphisms of rigid spaces such that $\varphi_i|_{X_i \cap X_j} \colon X_i \cap X_j \longrightarrow Y$ coincides with $\varphi_j|_{X_i \cap X_j} \colon X_i \cap X_j \longrightarrow Y$ for all $i, j \in I$. Then there is a unique morphism of rigid spaces $\varphi \colon X \longrightarrow Y$ satisfying $\varphi|_{X_i} = \varphi_i$ for all $i \in I$.*

Finally, we give the definition of the connectedness for rigid spaces.

Definition 1.10. *A rigid space X is called connected if there do not exist non-empty admissible open subspaces $X_1, X_2 \subset X$ such that $X_1 \cap X_2 = \varnothing$ and (X_1, X_2) is an admissible covering of X.*

2 An introduction to Formal Geometry

For simplicity, we only introduce the formal scheme over $\mathbb{O}$. Recall that $\mathbb{O}$ is the ring of integers of a completed non-archimedean valuation field $\mathbb{K}$. Assume we pick an element $\pi \in \mathbb{K}$ such that $|\pi| < 1$. If $\mathbb{O}$ is a DVR, we choose π to be the uniformizer.

2.1 Formal schemes

Let A be an $\mathbb{O}$-algebra that is complete and separated under the π-adic topology. Henceforth, unless otherwise specified, a "π-adic ring" refers to an $\mathbb{O}$-algebra of this type. Let $\operatorname{Spf} A$ be the collection of open prime ideals of A.

Lemma 2.1.1. *As sets,* $\operatorname{Spf} A = \operatorname{Spec}(A/(\pi))$.

Proof. By definition, $\operatorname{Spf} A = \{\mathfrak{p} \in \operatorname{Spec} A \mid \mathfrak{p} \text{ is open }\}$. If $\mathfrak{p}$ is open, then $(\pi^n) \subset \mathfrak{p}$ for some positive integer n. Thus, $\pi \in \mathfrak{p}$. Conversely, if $\pi \in \mathfrak{p}$, then $a + (\pi) \subset \mathfrak{p}$ for all $a \in \mathfrak{p}$, which means that $\mathfrak{p}$ is open.

Therefore, $\operatorname{Spf} A = \{\mathfrak{p} \in \operatorname{Spec} A \mid \pi \in \mathfrak{p}\} = \operatorname{Spec}(A/(\pi))$. □

We endow $\operatorname{Spf} A$ with the topology of $\operatorname{Spec}(A/(\pi))$. As usual, let $D(f)$ for $f \in A$ be the collection of open prime ideals that do not contain f, and we call them principal open sets. Then these principal open sets form a basis for the topology.

Then we will give the ringed space structure of $\operatorname{Spf} A$. Let $\mathscr{O}$ be the presheaf defined by sending $D(f)$, $f \in A$, to $A\langle f^{-1}\rangle$, the π-adic completion of $A[f^{-1}]$. It is easy to show that

$$A\langle f^{-1}\rangle = \varprojlim A[f^{-1}]/(\pi^n).$$

We will next show that $\mathscr{O}$ is actually a sheaf. For any covering $\{D(f_i)\}_i$ of $D(f)$, we will show that the following sequence is exact.

$$A\langle f^{-1}\rangle \longrightarrow \prod_i A\langle f_i^{-1}\rangle \rightrightarrows \prod_{i,j} A\langle (f_i f_j)^{-1}\rangle$$

Since $\operatorname{Spec}(A/(\pi^n))$ is a scheme for any $n \in \mathbb{N}$, we have the following exact sequence.

$$A[f^{-1}]/(\pi^n) \longrightarrow \prod_i A[f_i^{-1}]/(\pi^n) \rightrightarrows \prod_{i,j} A[(f_i f_j)^{-1}]/(\pi^n)$$

Passing through inverse limit, a left exact functor, we get our desired exact sequence; note that taking the inverse limit commutes with products.

Proposition 2.1.2. *The stalks of $\mathscr{O}$ are local rings.*

Definition 2.1. *For a π-adic ring A, set $X = Spf\ A$ and let $\mathscr{O}_X$ be the sheaf of topological rings we have constructed above. Then the locally ringed space $(X, \mathscr{O}_X)$ (where "ringed" has to be understood in the sense of topological rings) is called the affine formal scheme of A. It is again denoted by Spf A.*

Definition 2.2. *A formal scheme is a locally topologically ringed space $(X, \mathscr{O}_X)$ such that each point $x \in X$ admits an open neighborhood U where $(U, \mathscr{O}_X|_U)$ is isomorphic to an affine formal scheme Spf A, as constructed above.*

Definition 2.3. *A π-adic ring, A, is said to be of finite type if there is a surjective morphism from $\mathbb{O}\langle\zeta_1, \cdots, \zeta_n\rangle$, the π-adic completion of $\mathbb{O}[\zeta_1, \cdots, \zeta_n]$, to A.*

We call a formal scheme $(X, \mathscr{O}_X)$ is locally of finite type, if there is an open affine covering $\{U_i\}_i$ such that $(U_i, \mathscr{O}_X|_{U_i})$ is isomorphic to $\operatorname{Spf} A_i$ *where A_i is of finite type.*

Example 1. *Let X be an $\mathbb{O}$-scheme and $Y \subset X$ the special fibre of X. Then consider the sheaf $\mathcal{O}_Y$ obtained by restricting the inverse limit $\varprojlim \mathcal{O}_X/(\pi^n)$ to Y. It follows that $(Y, \mathcal{O}_Y)$ is a locally topologically ringed space, the desired formal completion of X along the special fibre. Locally, the construction looks as follows: Let $X = Spec\ A$. Then*

$$(Y, \mathcal{O}_Y) = Spf\left(\varprojlim A/(\pi^n)\right) = Spf\ \hat{A},$$

where $\hat{A}$ is the π-adic completion of A.

2.2 The geometry of formal schemes

We will introduce some geometry of formal schemes.

The open immersion of formal schemes are in the sense of open immersion of locally topologically ringed spaces.

We also have similar propositions about gluing formal schemes and gluing morphisms. We omit the details.

Fibre products exist in the category of formal schemes. We provide a description for the affine case in the following lemma; the general case is then obtained by gluing.

Lemma 2.2.1. *The fibre product* $\operatorname{Spf} A \times_{\operatorname{Spf} R} \operatorname{Spf} B = \operatorname{Spf}(\widehat{A \otimes_R B})$, *where $\widehat{A \otimes_R B}$ is the π-adic completion of $A \otimes_R B$.*

As usual, we omit the proof.

For any formal scheme X, we call $X \times_{\mathbb{O}} \operatorname{Spf}(\mathbb{O}/\pi\mathbb{O})$ the special fibre of X, which is also a scheme. It is easy to show that X is homeomorphic to its special fibre as topological spaces.

Then we briefly introduce the reducedness and the separatedness for formal schemes.

Definition 2.4. *An affine formal scheme* $\operatorname{Spf} A$ *is called reduced if A is reduced. A formal scheme X is called reduced if X admits an affine open covering $\{U_i\}_i$ such that U_i is reduced.*

Definition 2.5. *A formal scheme X is called separated if the diagonal morphism $\Delta : X \to X \times_{\mathbb{O}} X$ is a closed morphism.*

Proposition 2.2.2.

1. *Ler A be a π-adic ring. If π is a nonzero divisor in A, then A is reduced if $A/\pi A$ is reduced.*
2. *Let X be a formal scheme which is locally of finite type, then X is separated if the special fibre of X is separated as a scheme.*

We will conclude this section with a statement on the uniqueness of extension of morphisms from reduced formal schemes to separated schemes.

Proposition 2.2.3. *Let X be a reduced formal scheme and Y a separated formal scheme. Let $f, g : X \to Y$ be two morphisms. If $f|_U = g|_U$ for some dense open subset $U \subseteq X$, then $f = g$.*

The proof is analogous to the corresponding result in the theory of schemes.

2.3 Taking the generic fibre

Let A be a π-adic ring of finite type. Then $A \otimes_{\mathbb{O}} \mathbb{K}$ is an affinoid algebra, which yields an associated affinoid space $\mathrm{Sp}(A \otimes_{\mathbb{O}} \mathbb{K})$. For a general formal scheme X locally of finite type, we obtain the associated rigid space by gluing these local affinoid pieces.

Theorem 2.3.1. *Taking the generic fibre induces a functor from the category of formal schemes of locally finite type to the category of rigid spaces. We denote it by* $\mathfrak{rig}$.

Proposition 2.3.2. *The functor* $\mathfrak{rig}$ *sends open immersions to open immersions.*

Proposition 2.3.3. *The functor* $\mathfrak{rig}$ *sends connected formal schemes to connected rigid spaces.*

Part II
The p-adic upper half plane

Let K be a local non-archimedean field, $\mathcal{O}$ the ring of integers of K and π a uniformiser of $\mathcal{O}$. Let $k = \mathcal{O}/\pi\mathcal{O}$ be the residue field, p its characteristic, and q its order. We write $\mathbf{C}$ for the completion of the algebraic closure of K and $|\cdot|$ for the norm on $\mathbf{C}$ normalized such that $|\pi| = q^{-1}$. The valuation v is given by: $v(x) = \log_q |x|$. Let $\mathcal{O}_{\mathbf{C}}$ be the ring of integers of $\mathbf{C}$.

3 The building of $\mathrm{PGL}_2(K)$

3.1 The tree I

A lattice M of K^2 is an $\mathcal{O}$-submodule which is free of rank 2. Two lattices M and M' are homothetic if there exists $\lambda \in K^\times$ such that $M' = \lambda M$. We write S for the collection of homothety classes of lattices and $[M]$ for the class of the lattice M.

The building of $\mathrm{PGL}(2, K)$ is the graph I with vertices given by S, and where $s = [M]$ is joined to s' by an edge if and only if there exists a representative lattice M' for s' such that $\pi M \subsetneq M' \subsetneq M$.

Lemma 3.1.1. *The graph I is a homogeneous tree of degree $q + 1$.*

Proof. The degree assertion follows from the fact that the edges leaving a given vertex $[M]$ correspond to the distinct lattices M' satisfying the adjacency relation

$$\pi M \subsetneq M' \subsetneq M,$$

and these in turn are in bijection with the one-dimensional $\mathcal{O}/\pi\mathcal{O}$-subspaces in the two-dimensional k-vector space $M/\pi M$. There are $q + 1$ such subspaces, thus there are $q + 1$ adjacent vertices.

I is clearly connected.

Suppose that I is not a tree. A cycle in I would be minimally represented by a chain of lattices

$$M' \subsetneq M_d \subsetneq M_{d-1} \subsetneq \dots M_1 \subsetneq M,$$

where $M' = \pi^r M$ for some positive integer r and where none of the intermediate lattices are equivalent. Because M/M' is not a cyclic $\mathcal{O}$-module, there is a smallest i such that M/M_i is cyclic but M/M_{i+1} is not. It follows that M_{i-1}/M_{i+1} is a non-cyclic, length 2 $\mathcal{O}$-module, thus $M_{i+1} = \pi M_{i-1}$. This contradicts the minimality of the representation of the cycle. Thus, we conclude that I has no cycles. □

Remark 3.1.2. *We could easily deduce from this that the collection of vertices in I is countable.*

3.2 The geometric realization of I

The geometric realization of I, denoted by $\mathcal{I}_{\mathbb{R}}$, is a simplicial complex obtained by assigning a $[0, 1]$-structure to each edge of I. We shall show that $I_{\mathbb{R}}$ can be identified with the set of homothety classes of norms on the K-vector space K^2. Two norms $|\cdot|$ and $|\cdot|'$ are said to be proportional if there exists a constant $c \in \mathbb{R}_{>0}$ such that $|\cdot| = c|\cdot|'$.

We give some lemmas first.

Lemma 3.2.1. *Let M, M' be lattices in K^2 such that $\pi M \subsetneq M' \subsetneq M$, then we could find a basis for K^2, $\{e_1, e_2\}$, such that $M = \mathcal{O}e_1 \oplus \mathcal{O}e_2$ and $M' = \mathcal{O}e_1 \oplus \mathcal{O}\pi e_2$.*

Proof. We have the following commutative diagram.

$$\begin{array}{ccccc} 0 & \hookrightarrow & M/M' & \hookrightarrow & M/\pi M \\ \| & & \cong\| & & \|\cong \\ 0 & \hookrightarrow & k & \hookrightarrow & k^2 \end{array}$$

Thus, we could choose an $\mathcal{O}$-basis for M, $\{e_1, e_2\}$, such that e_1 is a k-basis for $M'/\pi M$. Then $M = \mathcal{O}e_1 \oplus \mathcal{O}e_2$ and $M' = \mathcal{O}e_1 \oplus \mathcal{O}\pi e_2$. □

Lemma 3.2.2. *Let $|\cdot|$ be a norm on K^2, then $M_\lambda = \{v \in K^2 \mid |v| \leq \lambda\}$ is a lattice in K^2.*

Proof. Since every norm on K^2 is equivalent to each other, we may assume $|\cdot| \geq c\,|\cdot|_0$, where c is a positive number and $|\cdot|_0$ is the standard norm on K^2. (To be explicit, $|(a,b)|_0 = \max\{|a|, |b|\}$ for any $(a,b) \in K$.) Then M_λ is contained in $\{(a,b) \in K^2 \mid \max\{|a|_0, |b|_0\} \leq c^{-1}\cdot\lambda\}$ which is a finitely generated $\mathcal{O}$-module. Since $\mathcal{O}$ is a PID, M_λ is also finitely generated. It is clear that M is a torsion free $\mathcal{O}$-module and $M \otimes_{\mathcal{O}} K = K^2$. Then M is a free $\mathcal{O}$-module of rank 2; that is, it is a lattice. □

Proposition 3.2.3. *There is a one-one correspondence between the points of the geometric realization $I_{\mathbb{R}}$ of I and the proportionality classes of norms on the K-vector space K^2*

Proof. **Step 1:** Vertices.

To a vertex $s = [M]$ corresponds the class of the norm $|\cdot|_M$ such that the corresponding unit ball is M. To be explicit, if (e_1, e_2) is a basis for M and if $v = a_1e_1 + a_2e_2$, we define

$$|v|_M = \sup\{|a_1|, |a_2|\}.$$

It is clear that $|\cdot|_M$ is a well-defined norm and $M = \{v \in K^2 \mid |v|_M \leq 1\}$.

Step 2 Edges.

Let s and s' be two adjacent vertices represented by lattices M and M', respectively. We may assume $\pi M \subset M' \subset M$. By Lemma 3.2.1, there exists a basis (e_1, e_2) of M such that $(e_1, \pi e_2)$ is a basis for M'. For $v = a_1e_1 + a_2e_2$, the norms are given by

$$\begin{aligned} |v|_M &= \sup\{|a_1|, |a_2|\}, \\ |v|_{M'} &= \sup\{|a_1|, q|a_2|\}. \end{aligned}$$

To a point $x = (1-t)s + ts'$, with $0 < t < 1$, on the edge between s and s', corresponds the class of the norm $|\cdot|_t$ defined by

$$|v|_t = \sup |a_1|, q^t|a_2|.$$

One directly verifies that:

$$\begin{aligned} M &= \{v \in K^2 \mid |v|_t \leq \lambda\} \text{ for } q^t \leq \lambda < q, \\ M' &= \{v \in K^2 \mid |v|_t \leq \lambda\} \text{ for } 1 \leq \lambda < q^t. \end{aligned}$$

Step 3: From norms to the points in $I_{\mathbb{R}}$.

Let $|\cdot|$ be a norm on K^2. For real $\lambda > 0$, the collection $M_\lambda = \{v \in K^2 \mid |v| \leq \lambda\}$ is a lattice in K^2 by Lemma 3.2.2. We have $M_{\lambda'} \subset M_\lambda$ if $\lambda' \leq \lambda$ and $M_{q^{-1}\lambda} = \pi M_\lambda$, thus $[M_\lambda]$ takes at most two values in S as λ varies.

If $[M_\lambda] = s$ is constant, then $|\cdot|$ corresponds to s.

Otherwise $[M_\lambda]$ equals s or s' for two adjacent vertices of I. After possibly replacing $|\cdot|$ by a proportional norm, one has $[M_\lambda] = s$ for $q^t \leq \lambda < q$ and $[M_\lambda] = s'$ for $1 \leq \lambda < q^t$, with $0 < t < 1$. Then $|\cdot|$ corresponds with the point $x = (1-t)s + ts'$ of the edge joining s and s'.

□

4 The p-adic upper half plane (rigid geometry version)

4.1 The map λ

We write $\Omega(\mathbf{C}) = \mathbb{P}^1(\mathbf{C}) - \mathbb{P}^1(K)$. If we identify $\mathbb{P}^1(\mathbf{C})$ with the collection of $\mathbf{C}^\times$-homothety classes of nonzero K-linear maps of K^2 into $\mathbf{C}$, then $\mathbb{P}^1(K)$ corresponds to those maps with K-rank equal to one. Thus Ω is identified with the collection of $\mathbf{C}^\times$-homothety classes of K-linear injective maps of K^2 into $\mathbf{C}$.

By composing a K-linear injective map $z : K^2 \to \mathbf{C}$ with the norm on $\mathbf{C}$, one obtains a norm $|\cdot|_z$ on K^2:

$$|v|_z = |z(v)|, \text{ for } v \in K^2.$$

This defines a map $\lambda : \Omega \to I_{\mathbb{R}}$

$$\lambda(\text{class of } z) = \text{class of } |\cdot|_z.$$

4.2 The corresponding affinoid spaces

Let $s = [M]$ and $s' = [M']$ be two adjacent vertices of I and (e_1, e_2) a basis adapted to the pair, so that $(e_1, \pi e_2)$ is a basis of M'. Identify Ω with $\mathbf{C} - K$ and choose a representative z for a point of Ω such that $z(e_2) = 1$ and $z(e_1) = \zeta \in \mathbf{C} - K$. Then we have the following proposition.

Proposition 4.2.1. *We have*

$$\lambda^{-1}(s) = \{\zeta \in \mathbf{C} \mid |\zeta| \le 1\} - \bigcup_{a \in \mathcal{O}/\pi\mathcal{O}} \{\zeta \in \mathbf{C} \mid |\zeta - a| < 1\},$$

$$\lambda^{-1}(x) = \{\zeta \in \mathbf{C} \mid |\zeta| = q^{-t}\} \quad \textit{if } x = (1-t)s + ts',\ 0 < t < 1,$$

$$\lambda^{-1}(s') = \{\zeta \in \mathbf{C} \mid |\zeta| \le q^{-1}\} - \bigcup_{b \in \pi\mathcal{O}/\pi^2\mathcal{O}} \{\zeta \in \mathbf{C} \mid |\zeta - b| < q^{-1}\},$$

$$\lambda^{-1}([s, s']) = \{\zeta \in \mathbf{C} \mid |\zeta| \le 1\} - \bigcup_{a \in (\mathcal{O} - \pi\mathcal{O})/\pi\mathcal{O}} \{\zeta \in \mathbf{C} \mid |\zeta - a| < 1\}$$
$$- \bigcup_{b \in \pi\mathcal{O}/\pi^2\mathcal{O}} \{\zeta \in \mathbf{C} \mid |\zeta - b| < q^{-1}\}.$$

In other words $\lambda^{-1}(s)$ (resp. $\lambda^{-1}(s')$) is the closed disc of radius 1 (resp. q^{-1}) centered on 0, but with the q open discs of radius 1 (resp. q^{-1}) centered on the K-rational points of the disc removed, while $\lambda^{-1}(]s, s'[)$ is the open anulus of interior radius q^{-1} and exterior radius 1 centered on 0.

Proof. The norms $|\cdot|_z = |\zeta a_1 + a_2|$ and $|\cdot|_t = \max\{|a_1|, q^t|a_2|\}$ are proportional on K^2 if and only if $|\cdot|_z = q^{-t} \cdot |\cdot|_t$. This equality is satisfied if and only if it is so for $a_1 = 1$, which is to say if

$$(*) \quad |\zeta + a_2| = \max\{q^{-t}, |a_2|\} \text{ for } a_2 \in K.$$

If $0 < t < 1$, one has $|a_2| \neq q^{-t}$ for all $a_2 \in K$, so that $(*)$ is equivalent with $|\zeta| = q^{-t}$.

On the other hand, if $t = 0$, $(*)$ is equivalent with $|\zeta| = 1$ and $|\zeta + a_2| = 1$ for all $a_2 \in K$ such that $|a_2| = 1$, or $|\zeta| \le 1$ and $|\zeta - a| \ge 1$ for $a \in \mathcal{O}$.

Finally, if $t = 1$, $(*)$ is equivalent to $|\zeta| = q^{-1}$ and $|\zeta + a_2| = q^{-1}$ for all $a_2 \in K$ such that $|a_2| = q^{-1}$, or $|\zeta| \le q^{-1}$ and $|\zeta - b| \ge q^{-1}$ for $b \in \pi\mathcal{O}$. □

The collections $\lambda^{-1}(s)$, $\lambda^{-1}(s')$ and $\lambda^{-1}([s, s'])$ possess natural structures as rigid analytic spaces defined over K; they are affinoid subsets of $\mathbb{P}^1_K$, the rigid projective line, that is, each is a complement of a finite number of open disks in $\mathbb{P}^1_K$. Moreover $\lambda^{-1}(s)$ and $\lambda^{-1}(s')$ are opens of $\lambda^{-1}([s, s'])$.

More precisely, if we interpret $\mathbb{P}^1_K$ as $\mathbf{C} \cup \{\infty\}$, then $\{\zeta \in \mathbf{C} \mid |\zeta| \le 1\}$ is the disk $B = \operatorname{Sp} \mathcal{O}\langle\zeta\rangle$, the complement of the open disk $\{\zeta \in \mathbf{C} \mid |\zeta| > 1\} \cup \{\infty\}$. Then $\lambda^{-1}(s), \lambda^{-1}(s'), \lambda^{-1}([s, s'])$ are special laurent domains in B.

To be explicit,

$$\lambda^{-1}(s) = B((\zeta - a_i)^{-1} \mid 0 \le i \le q-1),$$
$$\lambda^{-1}(s') = B(\pi^{-1}\zeta,\ \pi(\zeta - \pi a_i)^{-1} \mid 1 \le i \le q-1),$$
$$\lambda^{-1}([s,s']) = B(\pi\zeta^{-1},\ \pi(\zeta - \pi a_i)^{-1}, (\zeta - a_i)^{-1} \mid 1 \le i \le q-1),$$

where $\{a_i\}_{0\le i\le q-1}$ is a representative of k in $\mathcal{O}$ and $\overline{a_0} = 0 \in k$.

For notational convenience, we do not distinguish between the set and its corresponding affinoid space. Actually, the set $\lambda^{-1}(s)$ (resp. $\lambda^{-1}([s,s'])$) is the $\mathbf{C}$-points of the affinoid space $\lambda^{-1}(s)$ (resp. $\lambda^{-1}([s,s'])$).

We will give more rigorous constructions of $\lambda^{-1}(s)$ and $\lambda^{-1}([s,s'])$. (They are the generic fibres of some special formal scheme.)

4.3 The rigid p-adic upper half plane

More generally, if T is a finite subtree of I, then $\lambda^{-1}(T)$ corresponds to an affinoid subdomain of $\mathbb{P}^1_K$. It is obtained by gluing the affinoid spaces $\lambda^{-1}([s,s'])$ along the common intersections $\lambda^{-1}(s)$, where $[s,s']$ ranges over the edges of T and s denotes the interior vertices of T. We call a vertex interior if its degree is not less than 2.

Thus, $\Omega = \bigcup \lambda^{-1}(T)$, for all finite subtrees T of I, has a natural structure as a rigid space defined over K; it is an open subspace of $\mathbb{P}^1_K$. Moreover, its $\mathbf{C}$-points are exactly $\Omega(\mathbf{C})$.

In this way, we can imagine Ω as a tubular neighborhood of $I_{\mathbb{R}}$.

4.4 * An alternative construction of the rigid p-adic upper half plane

This subsection is optional and may be skipped by the reader.

To construct Ω, we first define an admissible covering that determines its rigid structure. We define an increasing sequence of affinoid subdomains Ω_n^- of $\mathbb{P}^1_K$ for $n \ge 1$, along with a related collection of admissible open sets Ω_n, such that Ω is the union of these subdomains. Geometrically, Ω_n^- is obtained by removing from $\mathbb{P}^1_K$ a collection of disjoint open discs of increasingly smaller radii centered at the K-points.

We will describe this in a coordinate-free way.

Given $x \in \mathbb{P}^1_K(\mathbf{C})$, we could choose homogeneous coordinates $x = [x_0, x_1]$ for x that are unimodular, which means that both coordinates are integral, but at least one is not divisible by π. For a real number $r > 0$, let

$$B(x,r) = \{y \in \mathbb{P}^1(\mathbf{C}) : v(y_0x_1 - y_1x_0) \ge r\},$$

where we always take a unimodular representative $[y_0, y_1]$ of y. Also, we define

$$B^-(x,r) = \{y \in \mathbb{P}^1(\mathbf{C}) : v(y_0x_1 - y_1x_0) > r\}.$$

Lemma 4.4.1. *Let x and x' be two elements of $\mathbb{P}^1(K)$, and let n be a positive integer. Then $B(x,n) \cap B(x',n) \ne \varnothing$ if and only if $[x_0,x_1] \equiv \lambda[x_0',x_1']$ (mod π^n) for some unit $\lambda \in \mathcal{O}^*$.*

Proof. Suppose $y \in B(x,n) \cap B(x',n)$. Then we have the following equations.

$$v(x_1y_0 - y_1x_0) \ge n$$
$$v(x_1'y_0 - y_1x_0') \ge n.$$

Suppose for convenience that y_0 is a unit. Then we can conclude that

$$v(x_0'x_1 - x_1'x_0) \ge n.$$

This means that the vectors $[x_0,x_1]$ and $[x_0',x_1']$ are linearly dependent modulo π^n, which is the claim.

Conversely, if the vectors are linearly dependent modulo π^n, we can construct an element y in the intersection of the two sets by choosing a unimodular representative of the kernel of the matrix formed by x and x'. □

For each integer $n > 0$, let $\mathcal{P}_n$ be a set of representatives for the points of $\mathbb{P}^1(K)$ modulo π^n. Let $\Omega_n(\mathbf{C})$ be the set

$$\Omega_n := \mathbb{P}^1(\mathbf{C}) \setminus \bigcup_{x \in \mathcal{P}_n} B(x, n)$$

Let $\Omega_n(\mathbf{C})^- \subset \Omega_n(\mathbf{C})$ be the set

$$\Omega_n^-(\mathbf{C}) := \mathbf{P}^1(\mathbf{C}) \setminus \bigcup_{x \in \mathcal{P}_n} B^-(x, n-1)$$

Let

$$\Omega(\mathbf{C}) = \bigcup_n \Omega_n(\mathbf{C}) = \bigcup_n \Omega_n^-(\mathbf{C}).$$

We can make the sets $\Omega_n(\mathbf{C})$ and $\Omega_n^-(\mathbf{C})$ more explicit. Fix an integer $n \geq 1$. Then we can choose representatives for $\mathcal{P}_n$ as follows:

$[a_i, 1]$, where $\{a_i\}_{i=0}^{q^n-1}$ is a set of representatives in $\mathcal{O}$ for $\mathcal{O}/\pi^n\mathcal{O}$;

$[1, b_i]$, where $\{b_i\}_{i=0}^{q^{n-1}-1}$ is a set of representatives in $\pi\mathcal{O}$ for $\pi\mathcal{O}/\pi^n\mathcal{O}$.

Then it follows from the definitions that $\Omega_n(\mathbf{C})$ is the set of points $x \in \mathbb{P}^1(\mathbf{C})$ defined by the inequalities

$$\begin{aligned} v(z(x) - a_i) &< n \quad i = 0, \ldots, q^n - 1, \\ v\left(\frac{1}{z(x)} - b_i\right) &< n \quad i = 0, \ldots, q^{n-1} - 1. \end{aligned}$$

where $z([a, b]) = \frac{a}{b}$.

It will be useful later to have slightly different inequalities for the covering domains. In particular, it is easy to check that if $v(b) < n$, then

$$v(1/z - b) < n \iff v(z - 1/b) < n - 2v(b).$$

Consequently we can (choosing $b_0 = 0$) rewrite the system of inequalities defining $\Omega_n(\mathbf{C})$ as

$$\begin{aligned} v(z - a_i) &< n && \text{for } i = 0, \ldots, q^n - 1, \\ v(z - 1/b_i) &< n - 2v(b_i) && \text{for } i = 1, \ldots, q^{n-1} - 1, \\ v(z) &> -n, \end{aligned}$$

and, for $\Omega_n^-(\mathbf{C})$,

$$\begin{aligned} v(z - a_i) &\leq n - 1 && \text{for } i = 0, \ldots, q^n - 1, \\ v(z - 1/b_i) &\leq n - 1 - 2v(b_i) && \text{for } i = 1, \ldots, q^{n-1} - 1, \\ v(z) &\geq 1 - n. \end{aligned}$$

We denote the corresponding rigid space associated with $\Omega_n(\mathbf{C})$ (resp. $\Omega_n^-(\mathbf{C})$) by Ω_n (resp. Ω_n^-) which is a subspace of $\mathbb{P}^1_K$. And $\Omega_n(\mathbf{C})$ (resp. $\Omega_n^-(\mathbf{C})$) is exactly the $\mathbf{C}$-points of Ω_n (resp. Ω_n^-).

Proposition 4.4.2. *We could construct Ω to be an admissible open subdomain of $\mathbb{P}^1_K$, and the coverings of Ω by the families $\{\Omega_n\}_{n=1}^\infty$ and $\{\Omega_n^-\}_{n=1}^\infty$ are admissible coverings. In the latter case, the covering is by open affinoid domains.*

We omit the proof of this proposition.

5 The p-adic upper half plane (formal geometry version)

5.1 The projective line associated to a vertex in I

Let R be a commutative ring and M be a finitely generated free R-module.

Definition 5.1. *Let $Sym_R^n M$ be the n times symmetry tensor product of M over R. To be explicit,*

$$Sym_R^n M = (\bigotimes_{j=1}^{n} M)/\{x_1 \otimes \cdots \otimes x_i \cdots x_j \otimes x_n - x_1 \otimes \cdots \otimes x_j \cdots x_i \otimes x_n \mid x_i, x_j \in M, 1 \leq i \leq j \leq n\}.$$

Then we define

$$Sym_R M := \bigoplus_{n \geq 0} Sym_R^n M$$

to be a graded ring. Let $\mathbb{P}(M) := \mathrm{Proj}(Sym_{\mathcal{O}} M)$ be the associated projective line of M over R.

It is easy to show that $Sym_R M$ is isomorphic to the polynomial ring $R[X_1, ..., X_d]$ as graded ring, where $d = \mathrm{rank}\, M$.

When $R = \mathcal{O}$ and M is a lattice in K^2, $\mathbb{P}(M)$ is a projective line over $\mathcal{O}$ and the generic fibre of $\mathbb{P}(M)$ is $\mathbb{P}(M \otimes_{\mathcal{O}} K) \simeq \mathbb{P}^1_K$, the special fibre of $\mathbb{P}(M)$ is $\mathbb{P}(M \otimes_{\mathcal{O}} k) \simeq \mathbb{P}^1_k$.

Lemma 5.1.1. *If M and M' are homothetic, then $\mathbb{P}(M)$ and $\mathbb{P}(M')$ are isomorphic under a canonical $\mathcal{O}$-isomorphism.*

Proof. We may assume $M' = \pi M$, then it suffices to show that $Sym_{\mathcal{O}} M$ and $Sym_{\mathcal{O}}(\pi M)$ are isomorphic as $\mathcal{O}$ algebras. It is clear by definition. □

We may thus write $\mathbb{P}_s$ for the projective line $\mathbb{P}(M)$ over $\mathcal{O}$, where $s = [M]$ is the vertex of I corresponding to M. Let $\widehat{\mathbb{P}}_s$ be the completion of $\mathbb{P}_s$ along its special fibre.

From now on, we fix an $\mathcal{O}$ basis for M, write $M = \mathcal{O}e_1 \oplus \mathcal{O}e_2$. Then we may assume $\mathbb{P}(M) = \mathbb{P}^1_{\mathcal{O}}$ with generic fibre $\mathbb{P}^1_K$ and special fibre $\mathbb{P}^1_k$, and $Sym_{\mathcal{O}} M = \mathcal{O}[X, Y]$ where X (resp. Y) represents e_1 (resp. e_2).

For a scheme X and $x, y \in X$, we say that x specializes to y and y is specialized by x if $y \in \overline{\{x\}}$.

Lemma 5.1.2. *Let L be a finite field extension of K, and A be a finite generated $\mathcal{O}$ algebra contained in L. Then A is either a field or contained in $\mathcal{O}_L$, the ring of integers of L. In particular, the number of nonzero prime ideal of A is not greater than 1.*

Proof. If $A \subset \mathcal{O}_L$, then $\mathcal{O} \subset A \subset \mathcal{O}_L$. Since $\mathcal{O}_L$ is the integral closure of $\mathcal{O}$ in L, $\mathcal{O}_L$ is the integral closure of A. By the going-up theorem and the fact that $\mathcal{O}_L$ has only one nonzero prime ideal, A also has only one nonzero prime ideal.

Otherwise, if A is not contained in O_L, then there is an element a in A such that $|a| > 1$. Then for any nonzero $b \in A$, $|(a^m b)^{-1}| < 1$ for some $m \in \mathbb{N}$. Then $(a^m b)^{-1}$ is integral over $\mathcal{O}$. Then $(a^m b)^{-1}$ is contained in A. b is invertible in A. Therefore, A is a field, and hence A has no nonzero prime ideal. □

Proposition 5.1.3. *For every closed point $x \in \mathbb{P}^1_K$, there exists a unique closed point $y \in \mathbb{P}^1_k$ specialized by x in $\mathbb{P}^1_{\mathbb{O}}$.*

Proof. Let $\mathbb{P}^1_{\mathbb{O}} = \mathrm{Proj}(\mathbb{O}[X, Y])$. The closure $\overline{\{x\}}$ is covered by two affine patches:

$$(\overline{\{x\}} \cap D_+(X)) \cup (\overline{\{x\}} \cap D_+(Y)) = \overline{\{x\}}.$$

Let $\mathfrak{p} \subset \mathbb{O}[X, Y]$ be the homogeneous prime ideal of x. Since $\mathrm{ht}(\mathfrak{p}) = 1$ and $\pi \notin \mathfrak{p}$, and $\mathbb{O}[X, Y]$ is a UFD, we have $\mathfrak{p} = (F)$ for some irreducible primitive homogeneous polynomial $F(X, Y) = \sum_{i=0}^{n} a_i X^i Y^{n-i}$ with $a_i \in \mathbb{O}$. At least one $a_i \in \mathbb{O}^\times$.

On $D_+(X) \cong \mathrm{Spec}(\mathbb{O}[T])$, we consider $f(T) = F(T, 1)$. Let L/K be a finite extension containing a root α of f. Then $\overline{\{x\}} \cap D_+(X) \cong \mathrm{Spec}(\mathbb{O}[\alpha])$. Similarly, $\overline{\{x\}} \cap D_+(Y) \cong \mathrm{Spec}(\mathbb{O}[1/\alpha])$.

Case 1: $|\alpha| < 1$. The ring $\mathbb{O}[\alpha]$ is the valuation ring of L (or a subring thereof), and the unique point in the special fiber is $y = (\pi, T - \bar{\alpha})$.

Case 2: $|\alpha| > 1$. This reduces to Case 1 by considering $1/\alpha$ in the chart $D_+(Y)$.

Case 3: $|\alpha| = 1$. Both α and $1/\alpha$ are integral over $\mathbb{O}$. The intersection with the special fiber $\mathbb{P}^1_k$ is the point y defined by the homogeneous ideal $(\pi, F(X, Y))$.

In all cases, $\overline{\{x\}} \cap \mathbb{P}^1_k$ consists of a single point y. □

Remark 5.1.4. *We could use the valuation criterion to show the existence. Since x is closed in $\mathbb{P}^1_K$, the residue field of x is a finite field extension of K, write L. Consider the map*

$$\mathrm{Spec}(\mathcal{O}_L) \to \mathbb{P}^1_{\mathcal{O}}.$$

By the valuation criterion for properness, there exists a unique point $y \in \mathbb{P}^1_k$ such that the image of $\mathrm{Spec}(\mathcal{O}_L)$ *is $\{x, y\}$, then we could conclude that $y \in \overline{\{x\}}$.*

Specialization induces a map from $\mathbb{P}^1_K(L)$ (the L-points in $\mathbb{P}^1_K$) to $\mathbb{P}^1_k(k_L)$ (the k_L-points in $\mathbb{P}^1_k$) for every field extension L/K where L is a p-adic field with ring of integers $\mathcal{O}_L$ and residue field k_L. We denote this map by $\mathfrak{s}_L$, or simply $\mathfrak{s}$ if there is no confusion. It is well defined by the valuation criterion for properness, as $\mathbb{P}^1_{\mathcal{O}}$ is naturally proper over $\mathcal{O}$.

To be explicit, for any $\phi : \mathrm{Spec}(L) \to \mathbb{P}^1_K$, write $x = \phi(*)$, if $\tilde{\phi} : \mathrm{Spec}(O_L) \to \mathbb{P}^1_{\mathcal{O}}$ is the extension of ϕ, then $im(\tilde{\phi}) = \{x, \phi'(*)\}$ where $\phi' := \mathfrak{s}(\phi) \in \mathbb{P}^1_k(k_L)$. Then we have the following commutative diagram:

$$\begin{array}{ccc} K[T] & \xrightarrow{\phi^\#} & L \\ \uparrow & & \uparrow \\ \mathcal{O}[T] & \xrightarrow{\tilde{\phi}^\#} & \mathcal{O}_L \\ \downarrow & & \downarrow \\ k[T] & \xrightarrow{\phi'^\#} & k_L \end{array}$$

where we assume $im(\tilde{\phi}) \subset D_+(X)$ and $T = \frac{Y}{X}$.

We could give a method finding $\mathfrak{s}(\phi)$. Firstly, ϕ must factor through $D_+(X)$ or $D_+(Y)$, then we could find a ring homomorphism such that $|\phi^\#(T)| \leq 1$ where $T = \frac{X}{Y}$ or $\frac{Y}{X}$. Then we could restrict $\phi^\#$ to $\mathcal{O}[T]$ to obtain a homomorphism $\tilde{\phi} : \mathcal{O}[T] \to \mathcal{O}_L$. Finally, taking the quotient simultaneously yields a homomorphism $\phi' : k[T] \to k_L$ which is exactly $\mathfrak{s}(\phi)$.

Remark 5.1.5. *If $\phi^\#(T)$ is integral over K, then $x = \phi(*)$ will be a closed point in $\mathbb{P}^1_K$ and the image of $\mathfrak{s}(\phi)$ is exactly the unique point in $\mathbb{P}^1_k$ specialized by x by Proposition 5.1.3. Otherwise, $\phi(*)$ will be the generic point in $\mathbb{P}^1_K$, and $\mathfrak{s}(\phi)(*)$ can be any point in $\mathbb{P}^1_k$ depending on $\phi^\#$.*

Proposition 5.1.6. *The points of $\lambda^{-1}(s)$ are exactly the points of $\mathbb{P}^1_K(\mathbf{C})$ that do not specialize to the k-points of $\mathbb{P}^1_k$ (which is the special fibre of $\mathbb{P}_s$).*

Proof. Assume $\mathbb{P}^1_K(\mathbf{C}) = \mathbf{C} \cup \{\infty\}$, $\mathbb{P}^1_k(\bar{k}) = \bar{k} \cup \{\infty\}$. For all $z \in \mathbf{C} \cup \{\infty\}$,

$$\mathfrak{s}(z) = \begin{cases} \infty & \text{if } |z| > 1 \\ 0 & \text{if } |z| < 1 \\ \bar{z} \in \bar{k} & \text{if } |z| = 1 \end{cases}.$$

Therefore, $\mathbb{P}^1_K(\mathbf{C}) \setminus \mathfrak{s}^{-1}(\mathbb{P}^1_k(k)) = \lambda^{-1}(s)$ is clear. □

Proposition 5.1.7. *We have the following results:* $\mathbb{P}^1_K(\mathbf{C}) = \mathbb{P}_s(\mathcal{O}_\mathbf{C}) = \widehat{\mathbb{P}}_s(\mathcal{O}_\mathbf{C})$*, where* $\widehat{\mathbb{P}}_s$ *is the formal completion of* $\mathbb{P}_s$ *along its special fibre.*

Proof. The first equality holds for the valuation criterion for properness. The second equality holds for the following facts.

Fact 1. *Let* F *be a field and* S *be a finite set contained in* $\mathbb{P}^n_F$ *for some* $n \in \mathbb{N}_+$*. Then there exists a principal open subset* U *such that* $S \subset U$*.*

Fact 2. *Let* A, B *be two* O *algebras with completed* π*-adic topology. Then*

$$Hom_{\mathcal{O}}(\mathrm{Spf}(A), \mathrm{Spf}(B)) = Hom_{\mathcal{O},cont}(B, A),$$

where $Hom_{\mathcal{O},cont}(B, A)$ *means the collection of continuous* $\mathcal{O}$*-homomorphisms from* B *to* A*.*

Fact 3. *Let* A, B *be two* O *algebras, and* A *has completed* π*-adic topology. Let* $\hat{B}$ *be the completion of* B*. Then*

$$Hom_{\mathcal{O}cont}(\hat{B}, A) = Hom_{\mathcal{O}cont}(B, A) = Hom_{\mathcal{O}}(B, A).$$

Fact 4. *Let* A, B *be two* O *algebras.* $Hom_{\mathcal{O}}(\mathrm{Spec}(A), \mathrm{Spec}(B)) = Hom_{\mathcal{O}}(B, A)$*.*

□

We will then study the formal version of $\lambda^{-1}(s)$.

Lemma 5.1.8. $\mathbb{P}^1_k(k)$*, the* k*-points in* $\mathbb{P}^1_k$*, is a finite set, and the image of a* k*-point is a closed point in* $\mathbb{P}_s$*.*

Proof. It is clear. □

Definition 5.2. *Let* Ω_s *be the open subscheme of* $\mathbb{P}_s$ *which is the complement of the union of the* k*-points in* $\mathbb{P}^1_k$*.*

Ω_s is indeed open by Lemma 5.1.8.

Definition 5.3. *Let* $\widehat{\Omega}_s$ *be the formal completion of* Ω_s *along its special fibre.*

The underlying topological space of $\widehat{\Omega}_s$ is the complement of the set of all k-points in $\mathbb{P}^1_k$. Then $\widehat{\Omega}_s$ is an open formal subscheme of $\widehat{\mathbb{P}}_s$. Without changing $\widehat{\Omega}_s$, we could modify Ω_s to be the affine open subscheme with global section $\mathcal{O}[T][(T - a_i)^{-1} \mid i = 0, \cdots, q-1]$ where $T = \frac{X}{Y}$ and $\{a_i \in \mathcal{O} \mid i = 0, \cdots, q-1\}$ is a set of representatives of k in $\mathcal{O}$.

Corollary 5.1.9. *We have* $\widehat{\Omega}_s(\mathcal{O}_\mathbf{C}) = \lambda^{-1}(s)$*.*

Proof. Use the same identification in Proposition 5.1.7. A more essential reason for this is Proposition 5.1.10. □

Proposition 5.1.10. *The rigid analytic space* $\lambda^{-1}(s)$ *is the generic fibre of the formal scheme* $\widehat{\Omega}_s$*.*

Proof. We have

$$\begin{aligned}\widehat{\Omega}_s &= \varinjlim \mathrm{Spec}(\mathcal{O}[T][(T - a_i)^{-1} \mid i = 0, \cdots, q-1]/(\pi^n)) \\ &= \mathrm{Spf}(\mathcal{O}\langle T\rangle\langle (T - a_i)^{-1} \mid i = 0, \cdots, q-1\rangle),\end{aligned}$$

where $\{\overline{a_0}, \cdots, \overline{a_{q-1}}\} = k$.

Therefore, the generic fibre of $\widehat{\Omega}_s$ is

$$\mathrm{Sp}(K\langle T\rangle\langle (T - a_i)^{-1} \mid i = 0, \cdots, q-1\rangle) = \lambda^{-1}(s).$$

□

Remark 5.1.11. *We could also use this definition to give the rigid space structure of* $\lambda^{-1}(s)$*.*

5.2 The space associated to an edge in I

Let s' be a vertex in I adjacent to s. We may assume $s' = [M']$ and $s = [M]$ such that $\pi M \subsetneq M' \subsetneq M$. Then s' defines a unique k-point of the special fibre of $\mathbb{P}_s$, which is induced by $M \to M/M' \cong k$. We denote the closed point by $x_{s'} \in \mathbb{P}_s$.

Identify M/M' with k. Let $\varphi : M \to k$ be the corresponding homomorphism $\mathcal{O}$-module. It induces a k-module homomorphism $\bar{\varphi} : \bar{M} = M \otimes_{\mathcal{O}} k \to k$. Then $\bar{\varphi}$ induces a homomorphism between graded rings $Sym_k \bar{M}$ and $Sym_k k$, and hence we obtain a morphism

$$\varphi^* : \mathrm{Spec}(k) = \mathrm{Proj}(Sym_k k) \to \mathrm{Proj}(Sym_k \bar{M}) = \mathbb{P}_s \times_{\mathcal{O}} k = \mathbb{P}^1_k.$$

It is clear that the k-point defined above is independent of the choice of (M, M').

Definition 5.4. *Let $\mathbb{P}_{[s,s']}$ be the blowing up of $\mathbb{P}_s$ along the closed point $x_{s'}$.*

We will finally show that $\mathbb{P}_{[s,s']}$ is canonically isomorphic to $\mathbb{P}_{[s',s]}$, and its generic fibre is canonically isomorphic to $\mathbb{P}^1_K$, coinciding with the generic fibres of $\mathbb{P}_s$ and $\mathbb{P}_{s'}$.

For convenience, assume $M = \mathcal{O}e_1 \oplus \mathcal{O}e_2$ and $M' = \mathcal{O}e_1 \oplus \mathcal{O}\pi e_2$. Then we may identify $Sym_O M$ with $\mathcal{O}[X, Y]$ and $Sym_O M'$ with $\mathcal{O}[X, Y']$. Then the generic (resp. special) fibre of $\mathbb{P}_s = \mathbb{P}^1_{\mathcal{O}}$ is identified with $\mathbb{P}^1_K$ (resp. $\mathbb{P}^1_k$).

Lemma 5.2.1. *Under such identification, $x_{s'} = [0, 1] \in \mathbb{P}^1_k$.*

Proof. Consider the identification $\varphi : \mathbb{O}e_1 \oplus \mathbb{O}e_2 \to k$ given by $ae_1 + be_2 \mapsto \bar{b}$. This induces a map on the coordinate rings $\bar{\varphi} : k[X, Y] \to k[T]$ defined by $F(X, Y) \mapsto F(0, T)$, which in turn induces a morphism on the affine patch:

$$k[X/Y] \to k, \quad f(X/Y) \mapsto f(0).$$

It follows that the point $x_{s'}$ corresponds to the coordinates $[0, 1]$ in $\mathbb{P}^1_k$. □

Now, let us consider the blowing up of $\mathbb{P}^1_{\mathcal{O}}$ along $x_{s'} = [0, 1]$. Write $x_{s'} = x$ for simplicity.

We have the following Cartesian diagram:

$$\begin{array}{ccc} E & \longrightarrow & \mathrm{Bl}_x \mathbb{P}^1_{\mathcal{O}} \\ \downarrow & & \downarrow \beta \\ \{x\} & \hookrightarrow & \mathbb{P}^1_{\mathcal{O}} \end{array}$$

where $\beta : \mathrm{Bl}_x \mathbb{P}^1_{\mathcal{O}} \to \mathbb{P}^1_{\mathcal{O}}$ is the blowing up, and E is the scheme-theoretic preimage of x by β and is an effective Cartier divisor of $\mathrm{Bl}_x \mathbb{P}^1_{\mathcal{O}}$.

We know $\mathbb{P}^1_{\mathcal{O}} = D_+(X) \cup D_+(Y)$. Write $t = \frac{X}{Y}$, $U_0 = D_+(Y) = \mathrm{Spec}(\mathcal{O}[\frac{X}{Y}]) = \mathrm{Spec}(\mathcal{O}[t])$, and $U_1 = D_+(X) = \mathrm{Spec}(\mathcal{O}[\frac{Y}{X}]) = \mathrm{Spec}(\mathcal{O}[\frac{1}{t}])$.

Since $x \in U_0 \setminus U_1$, we have $\beta^{-1}(U_1) = U_1$ and $\beta^{-1}(U_0) = \mathrm{Bl}_x U_0$. The blow-up of U_0 at x is given by:

$$\mathrm{Bl}_x U_0 = \mathrm{Proj}\left(\bigoplus_{n \geq 0} I^n\right), \text{where } I = (\pi, t) \subset \mathcal{O}[t].$$

To avoid confusion, we write $I = (\Pi, T)$ as the degree one part of the graded ring $\bigoplus_{n\geq 0} I^n$. Then $\bigoplus_{n\geq 0} I^n = (\mathcal{O}[t])[\Pi, T]$ where $t\Pi = \pi T$. To be more clear,

$$\bigoplus_{n \geq 0} I^n = (\mathcal{O}[t])[\Pi, T] = (\mathcal{O}[t])[\tilde{\Pi}, \tilde{T}]/(t\tilde{\Pi} - \pi\tilde{T}),$$

where $\tilde{\Pi}, \tilde{T}$ are free variables.

Then $\mathrm{Bl}_x U_0 = D_+(\Pi) \cup D_+(T)$. Write $V_\Pi = D_+(\Pi) = \mathrm{Spec}(\mathcal{O}[t][\frac{T}{\Pi}])$, and $V_T = D_+(T) = \mathrm{Spec}(\mathcal{O}[t][\frac{\Pi}{T}]$. And we have the following identifications of affine charts:

$$\mathcal{O}[t]\left[\frac{\Pi}{T}\right] \cong \mathcal{O}[t,u]/(tu-\pi) \qquad \text{and} \qquad \mathcal{O}[t]\left[\frac{T}{\Pi}\right] \cong \mathcal{O}[t,v]/(\pi v - t) = \mathcal{O}[v],$$
$$t \mapsto t, \qquad\qquad t \mapsto t,$$
$$\frac{\Pi}{T} \mapsto u; \qquad\qquad \frac{T}{\Pi} \mapsto v.$$

Therefore, we obtain an affine open covering of $\mathrm{Bl}_x \mathbb{P}^1_{\mathcal{O}}$, $\mathrm{Bl}_x \mathbb{P}^1_{\mathcal{O}} = U_1 \cup V_T \cup V_\Pi$.

Remark 5.2.2. *We will find that good notation really gives us insight. In this case, we could figure out how to glue these open subsets to obtain* $\mathrm{Bl}_x \mathbb{P}^1_{\mathcal{O}}$.

Then, we will give explicit descriptions of U_1, V_T and V_Π by analyzing their generic and special fibres, the specialization relations between these fibres, and their gluing.

We fix a field extension L/K where L is a p-adic field with ring of integers $\mathcal{O}_L$ and residue field k_L.

U_1 is the most easy, it is just $\mathbb{A}_{\mathcal{O}}$, a disjoint union of $\mathbb{A}^1_k$ and $\mathbb{A}^1_K$.

The specialization map is given by the following commutative diagram:

$$\begin{array}{ccc} K[t_1] & \longrightarrow & L \\ \uparrow & & \uparrow \\ \mathcal{O}[t_1] & \longrightarrow & \mathcal{O}_L \\ \downarrow & & \downarrow \\ k[t_1] & \longrightarrow & k_L \end{array}$$

where t_1 is actually $\frac{1}{t}$.

For V_Π, things are almost the same as U_1.

For V_T, its generic fibre is

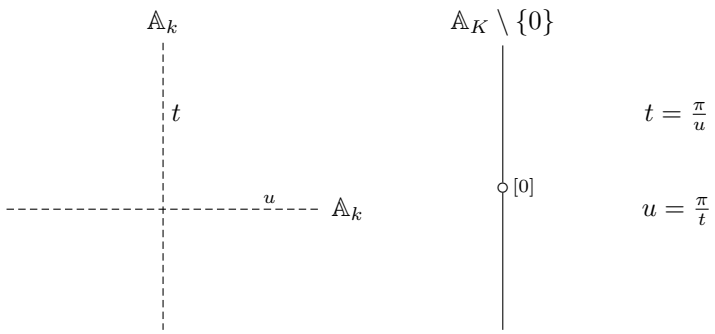
$$\mathrm{Spec}(K[t,u]/(tu-\pi)) = \mathrm{Spec}(K[t,\frac{1}{t}]) = \mathbb{A}_K \setminus \{0\}$$

parameterized by t, and its special fibre is

$$\mathrm{Spec}(\mathcal{O}[t,u]/(tu-\pi,\pi)) = \mathrm{Spec}(k[t,u]/(tu)) = \mathbb{A}_k \cup \mathbb{A}_k$$

which is a union of two affine lines parameterized by t and u intersecting at the "origin", or the union of the t-axis and the u-axis in $\mathbb{A}^2_k$.

We give a graph of V_T:



The specialization map is given by the following commutative diagram:

$$\begin{array}{ccc} K[t,\frac{1}{t}] = K[t,\frac{\pi}{t}] & \longrightarrow & L \\ \uparrow & & \uparrow \\ \mathcal{O}[t,u]/(tu-\pi) & \longrightarrow & \mathcal{O}_L \\ \downarrow & & \downarrow \\ k[t,u]/(t,u) & \longrightarrow & k_L \end{array}$$

If we take $L = \mathbf{C}$, then we could give a more specific description of the specialization map. We still denote it by $\mathfrak{s}$.

Example 2. *We have* $\mathfrak{s} : (V_T)_K(\mathbf{C}) \to (V_T)_k(\bar{k})$ *where* $(V_T)_K(\mathbf{C}) = \mathbf{C}$ *and* $(V_T)_k(\bar{k}) = \{(t,u) \in \bar{k} \mid t \cdot u = 0\}$.

$$\mathfrak{s}(a) = \begin{cases} (0,0) & |\pi| < |a| < 1 \\ (0,\bar{a}) & |a| = 1 \\ (\frac{\pi}{a},0) & |a| = |\pi| \\ non & otherwise \end{cases}$$

We want to describe the process of gluing V_T and V_Π. It suffices to give the isomorphism between their overlaps.

In V_T, the overlap is $\mathrm{Spec}(\mathcal{O}[t,u,\frac{1}{u}]/(tu-\pi)) = \mathrm{Spec}(\mathcal{O}[u,\frac{1}{u}])$.

Tn V_Π, the overlap is $\mathrm{Spec}(\mathcal{O}[v,\frac{1}{v}])$.

The relation is shown in the following diagram.

$$u \longmapsto \frac{1}{v}$$

$$\begin{array}{ccc} \pi \cdot \frac{1}{u} & & \\ \uparrow & & \\ t & & \end{array} \qquad \begin{array}{ccc} \mathcal{O}[u,\frac{1}{u}] & \xrightarrow{\cong} & \mathcal{O}[v,\frac{1}{v}] \\ \uparrow & & \uparrow \\ \mathcal{O}[t,u]/(tu-\pi) & & \mathcal{O}[v] \end{array}$$

More geometrically, the generic fibre of $\mathrm{Bl}_{\mathcal{O}} U_0$ is an affine line parameterized by t, and its special fibre is a reducible k-scheme consisting of an affine line parameterized by t and a projective line parameterized by u. These two components intersect transversely at a single point (t=0 and u=0).

The diagram for U_1 and V_T is shown below.

$$t \longmapsto \frac{1}{t_1}$$

$$\begin{array}{ccc} \pi \cdot \frac{1}{t} & & \\ \uparrow & & \\ u & & \end{array} \qquad \begin{array}{ccc} \mathcal{O}[t,\frac{1}{t}] & \xrightarrow{\cong} & \mathcal{O}[t_1,\frac{1}{t_1}] \\ \uparrow & & \uparrow \\ \mathcal{O}[t,u]/(tu-\pi) & & \mathcal{O}[t_1] \end{array}$$

Lastly, the diagram for U_1 and V_Π is shown below.

$$v \longmapsto \frac{t_1}{\pi}$$

$$\begin{array}{ccc} K[v] & \xrightarrow{\cong} & K[t_1] \\ \uparrow & & \uparrow \\ \mathcal{O}[v] & & \mathcal{O}[t_1] \end{array}$$

We now could give a complete description of $\mathrm{Bl}_x \mathbb{P}^1_{\mathcal{O}}$. Its generic fibre is an projective line parameterized simultaneously by t and u with the identification $t \cdot u = \pi$, and its special fibre is a reducible k-scheme consisting of two projective lines parameterized by t and u respectively. These two components intersect transversely at a single point (t=0 and u=0).

Applying the method described in Example 2 yields the following results:

Example 3. *We have an explicit description for the secialization map* $\mathfrak{s} : (\mathrm{Bl}_x \mathbb{P}^1_{\mathcal{O}})_K(\mathbf{C}) \to (\mathrm{Bl}_x \mathbb{P}^1_{\mathcal{O}})_k(\bar{k})$, *where* $(\mathrm{Bl}_x \mathbb{P}^1_{\mathcal{O}})_K(\mathbf{C}) = \mathbf{C} \sqcup \{\infty\}$ *and* $(\mathrm{Bl}_x \mathbb{P}^1_{\mathcal{O}})_k(\bar{k}) = \{(t,u) \in \bar{k} \sqcup \{\infty\} \mid t \text{ or } u = 0\}$. *For* $a \in \mathbf{C} \sqcup \{\infty\}$,

$$\mathfrak{s}(a) = \begin{cases} (0,0) & |\pi| < |a| < 1 \\ (0,\bar{a}) & |a| = 1 \\ (\overline{\frac{\pi}{a}},0) & |a| = |\pi| \\ (\infty,0) & |a| < |\pi| \\ (0,\infty) & |a| > 1 \text{ or } a = \infty \end{cases}.$$

Remark 5.2.3. *Identifying the generic fibre of* $\mathbb{P}_{[s,s']}$ *with* $\mathbb{P}^1_K$ *induces a bijection* j *from* $\lambda^{-1}([s,s'])$ *to those* $\mathbf{C}$*-points of* $\mathbb{P}_{[s,s']}$ *that do not specialize to the* k*-points of the special fibre except for the singular point. In particular,* $j(\lambda^{-1}(]s,s'[))$ *is those* $\mathbf{C}$*-points of* $\mathbb{P}_{[s,s']}$ *that speclialize to the singular* k*-point.*

It is time to show that $\mathbb{P}_{[s,s']}$ is canonically isomorphic to $\mathbb{P}_{[s',s]}$.

Proposition 5.2.4. $\mathbb{P}_{[s,s']}$ *is canonically isomorphic to* $\mathbb{P}_{[s',s]}$.

Proof. $\mathbb{P}_{[s',s]}$ is the blowing up of $\mathbb{P}_{s'}$ along its closed point defined by s. As usual, we assume $M = \mathcal{O}e_1 \oplus \mathcal{O}e_2$ ($M \sim \pi M = \mathcal{O}\pi e_1 \oplus \mathcal{O}\pi e_2$) and $M' = \mathcal{O}e_1 \oplus \mathcal{O}\pi e_2$, then $\mathbb{P}_{s'} = \mathrm{Proj}(\mathcal{O}[X,Y'])$ and $\mathbb{P}_{[s',s]} = \mathrm{Bl}_{x_s} \mathbb{P}_{s'}$ where $x_s = [1,0] \in \mathbb{P}^1_k \subset \mathbb{P}_{s'}$. For simplicity, write $x' = x_s$.

Denote $\frac{Y'}{X}$ by t'. Similar to the previous construction, we obtain an affine open covering of $\mathrm{Bl}_{x'} \mathbb{P}_{s'}$ consisting of the charts $U'_1, V'_{T'}$ and V'_Π. And

$$U'_1 = \mathrm{Spec}(\mathcal{O}[t'_1]),\ V'_{T'} = \mathrm{Spec}(\mathcal{O}[t',u']/(t'u' - \pi)) \text{ and } V'_\Pi = \mathrm{Spec}(\mathcal{O}[v']),$$

with the relations that $t'_1 = \frac{1}{t'},\ v' = \frac{1}{u'},\ u' = \frac{\pi}{t'}$ and $t' = \frac{\pi}{t}$.

To be explicit, we have the following isomorphisms:

$$V_T = \mathrm{Spec}(\mathcal{O}[t,u]/(tu - \pi)) \xrightarrow{\cong} V'_{T'} = \mathrm{Spec}(\mathcal{O}[t',u']/(t'u' - \pi))$$

$$\mathcal{O}[t,u]/(tu - \pi) \xleftarrow{\cong} \mathcal{O}[t',u']/(t'u' - \pi)$$

$$\begin{array}{ccc} u & \longleftarrow\!\shortmid & t' \\ t & \longleftarrow\!\shortmid & u' \end{array}$$

$$U_1 = \mathrm{Spec}(\mathcal{O}[t_1]) \xrightarrow{\cong} V'_\Pi = \mathrm{Spec}(\mathcal{O}[v'])$$

$$\mathcal{O}[t_1] \xleftarrow{\cong} \mathcal{O}[v']$$

$$t_1 \longleftarrow\!\shortmid v'$$

$$V_\Pi = \mathrm{Spec}(\mathcal{O}[v]) \xrightarrow{\cong} U_1' = \mathrm{Spec}(\mathcal{O}[t_1'])$$

$$\mathcal{O}[v] \xleftarrow{\cong} \mathcal{O}[t_1']$$

$$v \longleftarrow\!\shortmid\ t_1'$$

The coincidence in transitive maps is natural. Then we have canonically $\mathbb{P}_{[s,s']} \cong \mathbb{P}_{[s',s]}$. □

From now on, we will not distinguish $\mathbb{P}_{[s,s']}$ and $\mathbb{P}_{[s',s]}$.

We will then study the formal version of $\lambda^{-1}([s,s'])$.

Definition 5.5. *Let $\Omega_{[s,s']}$ be the open subscheme of $\mathbb{P}_{[s,s']}$ which is the complement of the (finite many) k-points of the special fibre except for the singular point.*

Definition 5.6. *Let $\widehat{\Omega}_{[s,s']}$ be the formal completion of $\Omega_{[s,s']}$ along its special fibre.*

Corollary 5.2.5. *By Remark 5.2.3, we could deduce that there is a bijection from $\lambda^{-1}([s,s'])$ to $\widehat{\Omega}_{[s,s']}(\mathcal{O}_{\mathbf{C}})$.*

Proof. Using the same argument in Corollary 5.1.9. □

Similar to Proposition 5.1.10, we have the following proposition:

Proposition 5.2.6. *The rigid analytic space $\lambda^{-1}([s,s'])$ is the generic fibre of the formal scheme $\widehat{\Omega}_{[s,s']}$.*

Proof. We could modify $\Omega_{[s,s']}$ with $\widehat{\Omega}_{[s,s']}$ unchanged. Thus, we could remove the points $[0,1]$ and $[1,0]$ in $\mathbb{P}^1_K \subset \Omega_{[s,s']}$. Then we could view $\Omega_{[s,s']}$ as an open subscheme in V_T. Moreover, we could let $\Omega_{[s,s']}$ be the following open subscheme of V_T:

$$\mathrm{Spec}\left((\mathcal{O}[t,u]/(tu-\pi))[(t-a_i)^{-1},(u-a_i)^{-1} \mid i=1,\dots,q-1]\right)$$

where $a_i \in \mathcal{O}$, $\{\overline{a_0},\dots,\overline{a_{q-1}}\} = k$ and $\overline{a_0} = \bar{0}$.

Then

$$\begin{aligned}\widehat{\Omega}_{[s,s']} &= \varinjlim \mathrm{Spec}\left((\mathcal{O}[t,u]/(tu-\pi))[(t-a_i)^{-1},(u-a_i)^{-1} \mid i=1,\dots,q-1]/(\pi^n)\right)\\ &= \mathrm{Spf}\left(\mathcal{O}\langle t,u\rangle\langle(t-a_i)^{-1},(u-a_i)^{-1} \mid i=1,\dots,q-1\rangle/(tu-\pi)\right).\end{aligned}$$

Therefore, the generic fiber of $\widehat{\Omega}_{[s,s']}$ is

$$\begin{aligned}&\mathrm{Sp}\left(K\langle t,u\rangle\langle(t-a_i)^{-1},(u-a_i)^{-1} \mid i=1,\dots,q-1\rangle/(tu-\pi)\right)\\ &= \mathrm{Sp}\left(K\langle t\rangle\langle(\pi^{-1}\cdot t)^{-1},(t-a_i)^{-1},(1-\tfrac{a_i}{\pi}\cdot t)^{-1} \mid i=1,\dots,q-1\rangle\right) = \lambda^{-1}([s,s']).\end{aligned}$$

The first equality is not trivial. To show that, we shall prove that the following two affinoid K algebras are isomorphic to each other.

$$\begin{aligned}A &= K\langle t,u,\xi_i \mid 1\le i\le q-1\rangle/(tu-\pi,(\pi-a_it)\xi_i-\pi \mid 1\le i\le q-1)\\ B &= K\langle t,u,\zeta_i \mid 1\le i\le q-1\rangle/(tu-\pi,(u-a_i)\zeta_i-1 \mid 1\le i\le q-1)\end{aligned}$$

Let $\phi: A\to B$ and $\psi: B\to A$ be two morphisms such that $\phi(\xi_i) = u\cdot\zeta_i$ and $\psi(\zeta_i) = \frac{t}{\pi}\cdot\xi_i$ where $1\le i\le q-1$. It is easy to check that ϕ and ψ are well-defined and inverse to each other. Thus, $A\cong B$. □

5.3 The formal p-adic upper half plane

Finally, we will reveal the relations between $\widehat{\Omega}_s$ and $\widehat{\Omega}_{[s,s']}$, and construct the formal p-adic upper half plane. Its generic fibre is the rigid p-adic upper half plane we constructed before.

Recall that we have the following cartesian diagram:

$$\begin{array}{ccc} E & \longrightarrow & \mathbb{P}_{[s,s']} \\ \downarrow & & \downarrow \beta \\ \{x_{s'}\} & \hookrightarrow & \mathbb{P}_s \end{array}$$

and $\beta|_U$ is an isomorphism from $U := \mathbb{P}_{[s,s']} \setminus E$ to $\mathbb{P}_s \setminus \{x_{s'}\}$.

Since Ω_s is contained in $\mathbb{P}_s \setminus \{x_{s'}\}$, we gain an isomorphism between Ω_s and its preimage by β. Since β is continuous, $\beta^{-1}(\Omega_s)$ is an open subscheme of $\mathbb{P}_{[s,s']}$. It is clear that $\beta^{-1}(\Omega_s)$ is contained in $\Omega_{[s,s']}$ by directly checking the special fibre and the generic fibre respectively; you will find that Ω_s could be obtained from $\mathbb{P}_{[s,s']}$ by taking the complement of the u-axis and the k-points of its special fibre. Therefore, we obtain an open immersion $i_{s,s'} : \Omega_s \to \Omega_{[s,s']}$. Analogously, we also have $i_{s',s} : \Omega_{s'} \to \Omega_{[s,s']}$.

We conclude our discussion above by the proposition below.

Proposition 5.3.1. *We have an open immersion $i_{s,s'} : \Omega_s \to \Omega_{[s,s']}$.*

Corollary 5.3.2. *$\widehat{\Omega}_s$ and $\widehat{\Omega}_{s'}$ are open formal subschemes of $\widehat{\Omega}_{[s,s']}$*

Proof. It is just a natural corollary of Proposition 5.3.1. □

Proposition 5.3.3. *Taking generic fibres, we could recover the open immersion $\lambda^{-1}(s) \hookrightarrow \lambda^{-1}([s,s'])$.*

Proof. The assertion follows immediately by Proposition 2.3.2, Proposition 5.1.10, and Proposition 5.2.6. □

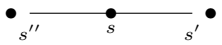
Corollary 5.3.4. *The intersection of $\widehat{\Omega}_s$ and $\widehat{\Omega}_{s'}$, as open subschemes of $\widehat{\Omega}_{[s,s']}$, is the generic point.*

Proof. It follows from the observation that $\lambda^{-1}(s) \cap \lambda^{-1}(s') = \varnothing$ and the irreducibility of $\widehat{\Omega}_{[s,s']}$. Or one can check directly since we already know the images of $\widehat{\Omega}_s$ and $\widehat{\Omega}_{s'}$ in $\widehat{\Omega}_{[s,s']}$. □

The cases of a single vertex and a single edge have been thoroughly investigated.

More generally, if T is a finite subtree of I, we could obtain a formal scheme $\widehat{\Omega}_T$ by gluing the formal schemes $\widehat{\Omega}_{[s,s']}$ along the $\widehat{\Omega}_s$ for $[s,s']$ an edge of T and s an interior vertex of T (according to the incidence relations of T). The generic fibre of $\widehat{\Omega}_T$ is canonically identified with $\lambda^{-1}(T)$ by the construction of $\mathtt{rig}$ (which is the taking the generic fibre functor).

Example 4. *Let T be a tree as the following graph.*



Then $\widehat{\Omega}_T$ is the formal scheme obtained by gluing $\widehat{\Omega}_{[s'',s]}$ and $\widehat{\Omega}_{[s,s']}$ via $\widehat{\Omega}_s$.

If $T \subset T'$, then the open immersion $\hat{\Omega}_T \hookrightarrow \hat{\Omega}_{T'}$ induces the open immersion $\lambda^{-1}(T) \hookrightarrow \lambda^{-1}(T')$ on the generic fibers.

Constructing in this way, we could obtain a formal scheme

$$\widehat{\Omega} = \bigcup_T \widehat{\Omega}_T,$$

whose generic fiber is Ω; in particular, $\widehat{\Omega}(\mathcal{O}_{\mathbf{C}}) = \Omega$. Since any morphism $\mathrm{Spf}(\mathcal{O}_{\mathbf{C}}) \to \widehat{\Omega}$ must factor through an open immersion $\widehat{\Omega}_{[s,s']}$, and the rest follows from Corollary 5.2.5.

We will conclude this section with an insightful result.

Proposition 5.3.5. *The special fiber of $\widehat{\Omega}$ is a tree of projective lines over k intersecting at k-rational points, which is dual to the tree I.*

Proof. We know that The special fibre of $\widehat{\Omega}_s$ is the special fibre of Ω_s, $(\Omega_s)_k$; similarly, the special fibre of $\widehat{\Omega}_{[s,s']}$ is $(\Omega_{[s,s']})_k$. The special fibre of $\widehat{\Omega}_T$ is glued by $(\Omega_{[s,s']})_k$'s via $(\Omega_s)_k$'s where T is a finite subtree of I. We denote it by $(\Omega_T)_k$. Thus, the special fibre of $\widehat{\Omega}$ is glued up by $(\Omega_T)_k$'s. We denote it by $\widehat{\Omega}_k$.

Then $(\Omega_s)_k$ is $(\mathbb{P}_s)_k = \mathbb{P}^1_k$ with k-points removed. $(\Omega_{[s,s']})_k$ is $(\mathbb{P}_{[s,s']})_k$, a union of two projective lines over k, with smooth k-points removed. You could somehow image $(\Omega_{[s,s']})_k$ as a "union" of $(\Omega_s)_k$ and $(\Omega_{s'})_k$ added with a "k-point" corresponding to the edge $[s, s']$.

Any k-point on $\mathbb{P}^1_k$ corresponds to a unique vertex adjacent to s, since there are only $q+1$ many k-points. Let T_s be the subtree consisting of s and vertexes adjacent to s and the associated edges, then the closed closure of $(\Omega_s)_k \setminus \{\eta\}$ in $(\Omega_{T_s})_k$ is homeomorphic to $(\mathbb{P}_s)_k \setminus \{\eta\}$ where η is the generic point. Thus,

$$(\Omega_s)_k \subset \mathbb{P}_s \subset (\Omega_{T_s})_k \subset \widehat{\Omega}_k,$$

where we view $\mathbb{P}_s$ as a subspace of $(\Omega_{T_s})_k$. We view $\mathbb{P}_s$ as a subspace of $\widehat{\Omega}_k$ in the following argument.

It is clear that

$$(\Omega_{T_s})_k \subset \mathbb{P}_s \cup \bigcup_{s' \sim s} \mathbb{P}_{s'}, \text{ and } \mathbb{P}_s \cap \mathbb{P}_{s'} = \{\eta, x\},$$

where $x \in (\mathbb{P}_{[s,s']})_k$ is the k-point corresponding to the edge $[s, s']$. Thus,

$$\widehat{\Omega}_k = \bigcup_{s \in I} (\Omega_{T_s})_k = \bigcup_{s \in I} \mathbb{P}_s.$$

Therefore, the special fiber of $\widehat{\Omega}$ is indeed a tree of projective lines over k intersecting at k-rational points, and it is dual to the tree I. □

Remark 5.3.6. *Then it is easy to show that the special fibre of $\widehat{\Omega}$ is connected. Thus, Ω is a connected rigid space by Proposition 2.3.3.*

6 The functor of Deligne

Definition 6.1. *Let* **Compl** *be the category of $\mathcal{O}$-algebras that are separated and complete for the π-adic topology, and the morphisms are $\mathcal{O}$-homomorphisms. Note that $\mathcal{O}$-homomorphisms between $\mathcal{O}$-algebras are naturally continuous.*

We describe, following Deligne, the functors from **Compl** to **Set** which are represented by the formal schemes $\widehat{\Omega}_s$ and $\widehat{\Omega}_{[s,s']}$.

6.1 The functor represented by $\widehat{\Omega}_s$

Definition 6.2. *We write F_s for the functor which, to $R \in$ **Compl**, associates the collection of isomorphism classes of pairs $(\mathcal{L}, \alpha)$, where $\mathcal{L}$ is a free R-module of rank 1 and $\alpha : M \to \mathcal{L}$ is a homomorphism of $\mathcal{O}$-modules satisfying the condition:*

$$(*) \begin{cases} \text{for all } x \in \mathrm{Spec}(R/\pi R), \text{ the map} \\ \alpha(x) : M/\pi M \to \mathcal{L} \otimes_R k(x) \text{ is injective.} \end{cases}$$

Remark 6.1.1. *We say that the two pairs $(\mathcal{L}_1, \alpha_1)$, $(\mathcal{L}_2, \alpha_2)$ are isomorphic if there is a R-isomorphism, ϕ, making the following diagram commutes.*

$$\begin{array}{ccc} M & \xrightarrow{\alpha_1} & \mathcal{L}_1 \\ \| & & \downarrow \phi \\ M & \xrightarrow{\alpha_2} & \mathcal{L}_2 \end{array}$$

And $Iso_R(\mathcal{L}_1, \mathcal{L}_2) = R^*$, the multiplicative group of R.

For any $\varphi : R \to R'$, we have a morphism $F_s(R') \to F_s(R)$ given by

$$[(\alpha', \mathcal{L}')] \mapsto [(\varphi \circ \alpha', \mathcal{L})].$$

It is well-defined since $\varphi(R^*) \subseteq R'^*$. Thus, F_s is indeed a functor.

Lemma 6.1.2. *If $a \in R$ such that $a \notin \mathfrak{m}$ for any $\mathfrak{m} \in \mathrm{Spec}(R/\pi R)$, then $a \in R^*$.*

Proof. By the assumption, $(a, \pi) = (1)$, and hence $b \cdot a + c \cdot \pi = 1$ for some $b, c \in R$. Since R is complete,

$$(1 - c \cdot \pi)^{-1} = 1 + (c \cdot \pi) + (c \cdot \pi)^2 + \cdots$$

is invertible, then so does a. □

Assume $M = \mathcal{O}e_1 \oplus \mathcal{O}e_2$ and $Sym_{\mathcal{O}} M = \mathcal{O}[X, Y]$ where X represents e_1 and Y represents e_2. And $M/\pi M = ke_1 \oplus ke_2$.

Lemma 6.1.3. *The condition $(*)$ is equivalent to $\alpha(x)(e_1)$ and $\alpha(x)(e_2)$ are k linear independent for all $x \in \mathrm{Spec}(R/\pi R)$, where we view $\mathcal{L} \otimes_R k(x)$ as a k-vector space. Moreover, $\alpha(e_i)$ $(i = 1, 2)$ is a generator of $\mathcal{L}$ as a R-module.*

Proof. Since $M/\pi M = ke_1 \oplus ke_2$, the condition $(*)$ is equivalent to

$$\mathrm{rank}_k \begin{pmatrix} \alpha(x)(e_1) \\ \alpha(x)(e_2) \end{pmatrix} = 2.$$

Then we've done the first part. And the "moreover" part follows by Lemma6.1.2. □

Corollary 6.1.4. *We have an injection $\iota : F_s(R) \hookrightarrow Hom_{\mathcal{O}}(\mathcal{O}[\zeta, \eta]/(\zeta\eta - 1), R)$.*

Proof. We first give the construction of the map. For any class of diagrams in $F_s(R)$, we have the unique elements $\zeta, \eta \in R$ satisfying that $\alpha(e_1) = \zeta \cdot \alpha(e_2)$ and $\alpha(e_2) = \eta \cdot \alpha(e_1)$, by Lemma 6.1.3. They are independent of the choice of the representatives.

It remains to show $\zeta \cdot \eta = 1$. It follows from

$$\zeta\eta\alpha(e_1) = \zeta\alpha(e_2) = \alpha(e_1)$$

and Lemma 6.1.3 ($\alpha(e_1)$ is a generator). □

Remark 6.1.5. *For any $\phi \in im(\iota)$, we could pick a representative of $\iota^{-1}(\phi)$ as following.*

$$M \xrightarrow{\alpha} R$$

such that $\alpha(e_1) = \phi(\zeta), \alpha(e_2) = 1$. By an abuse of notation, we no longer distinguish between $\phi(\zeta)$ and ζ

We know that

$$\begin{aligned}\mathrm{Hom}_{\mathcal{O}}(\mathcal{O}[\zeta,\eta]/(\zeta\eta - 1), R) &= \mathrm{Hom}_{\mathcal{O}}(\mathcal{O}\langle\zeta, \zeta^{-1}\rangle, R) \\ &\cong \mathrm{Hom}_{\mathcal{O}}(\mathrm{Spf}\, R, \mathrm{Spf}\, \mathcal{O}\langle\zeta, \zeta^{-1}\rangle).\end{aligned}$$

Proposition 6.1.6. *The functor F_s is represented by the formal scheme $\widehat{\Omega}_s$.*

Proof. Via Corollary 6.1.4, we identify F_s with a subfunctor of the formal scheme $\mathrm{Spf}(\mathcal{O}\langle\zeta, \zeta^{-1}\rangle)$. It is actually the open subscheme of $\widehat{\mathbb{P}}_s$ which is the complement of the points at infinity and zero.

By Remark 6.1.5, the condition $(*)$ can be expressed in terms of the image $\bar{\zeta}$ in $R/\pi R$: for all $a \in k$, $\bar{\zeta} - a$ is invertible in $R/\pi R$. Thus, F_s is the subfunctor of $\widehat{\mathbb{P}}_s - (\{\bar{\zeta} = \infty\} \cup \{\bar{\zeta} = 0\})$ which represents the open which is the complement of the k-rational points of the special fiber $\mathrm{Spec}(k[\bar{\zeta}, \bar{\zeta}^{-1}])$, which is exactly $\widehat{\Omega}_s$.

□

Remark 6.1.7. *We give another more direct proof. By Lemma 6.1.2, $\zeta - a_i \in R^*$ for $0 \le i \le q-1$ where $\{a_i\}_i$ is a representative of k and $\overline{a_0} = 0$.*

Therefore,

$$\begin{aligned}F_s(R) &= \{\zeta \in R \mid (\zeta - a_i) \in R^*, 0 \le i \le q-1\} \\ &= Hom_{\mathcal{O}}(\mathcal{O}[\zeta][(\zeta - a_i)^{-1} \mid 0 \le i \le q-1], R) \\ &= Hom_{\mathcal{O}}(\mathcal{O}\langle\zeta\rangle\langle(\zeta - a_i)^{-1} \mid 0 \le i \le q-1\rangle, R) \\ &= Hom_{\mathcal{O}}(\mathrm{Spf}\, R, \widehat{\Omega}_s) \\ &= \widehat{\Omega}_s(R).\end{aligned}$$

6.2 The functor represented by $\widehat{\Omega}_{[s,s']}$

Definition 6.3. *We write $F_{[s,s']}$ for the functor which, to $R \in$ **Compl**, associates the collection of isomorphism classes of commutative diagrams:*

$$\begin{array}{ccccc} \pi M & \hookrightarrow & M' & \hookrightarrow & M \\ \downarrow{\scriptstyle \alpha/\pi} & & \downarrow{\scriptstyle \alpha'} & & \downarrow{\scriptstyle \alpha} \\ \mathcal{L} & \xrightarrow{c} & \mathcal{L}' & \xrightarrow{c'} & \mathcal{L} \end{array}$$

where $\mathcal{L}$ and $\mathcal{L}'$ are free R-modules of rank 1, α and α' are homomorphisms of $\mathcal{O}$-modules, c and c' are homomorphisms of R-modules, satisfying the condition:

$$(**)\begin{cases} \text{for all } x \in \mathrm{Spec}(R/\pi R), \text{ one has} \\ \ker(\alpha(x) : M/\pi M \to \mathcal{L} \otimes_R k(x)) \subset M'/\pi M, \\ \ker(\alpha'(x) : M'/\pi M' \to \mathcal{L}' \otimes_R k(x)) \subset \pi M/\pi M'. \end{cases}$$

Remark 6.2.1. *By almost the same reason in the previous subsection, we could deduce that $F_{[s,s']}$ is indeed a functor.*

Assume $M = \mathcal{O}e_1 \oplus \mathcal{O}e_2$ and $M' = \mathcal{O}e_1 \oplus \mathcal{O}\pi e_2$; assume $Sym_{\mathcal{O}}M = \mathcal{O}[X, Y]$ and $Sym_{\mathcal{O}}M' = \mathcal{O}[X, Y']$, where X represents e_1, Y represents e_2 and Y' represents πe_2. Moreover, we have the following relation.

$$\begin{array}{ccc} M/M' = ke_2 & & M'/\pi M = ke_1 \\ \uparrow\!\uparrow & & \uparrow\!\uparrow \\ M/\pi M = ke_1 \oplus ke_2 & & M'/\pi M' = ke_1 \oplus ke_2' \\ \uparrow\!\downarrow & & \uparrow\!\downarrow \\ M'/\pi M = ke_1 & & \pi M/\pi M' = ke_2' \end{array}$$

In the above diagram, e_2' should be viewed as πe_2.

Lemma 6.2.2. *The condition $(**)$ implies that $\alpha(e_2)$ generates $\mathcal{L}$ and $\alpha'(e_1)$ generates $\mathcal{L}'$.*

Proof. The condition $(**)$ implies that $\alpha(x)(e_2) \neq 0$ for all $x \in \mathrm{Spec}(R/\pi R)$. Thus, $\alpha(e_2) \notin \mathfrak{m}_x\mathcal{L}$ for all $x \in \mathrm{Spec}(R/\pi R)$, and hence $\alpha(e_2)$ generates $\mathcal{L}$ as a R-module. For $\alpha'(e_1)$, the reason is the same. □

Remark 6.2.3. *Since*

$$c' \circ c(\alpha(e_2)) = c' \circ c((\alpha/\pi)(\pi e_2)) = \alpha(\pi e_2) = \pi\alpha(e_2),$$

we have $c' \circ c = \pi$ in each diagram.

Corollary 6.2.4. *We have an injection*

$$\iota : F_{[s,s']}(R) \hookrightarrow Hom_{\mathcal{O}}(\mathcal{O}[\zeta, \eta]/(\zeta\eta - \pi), R).$$

Proof. We first give the construction of the map. For any class of diagrams in $F_{[s,s']}(R)$, we have the unique elements $\zeta, \eta \in R$ satisfying that $\alpha(e_1) = \zeta \cdot \alpha(e_2)$ and $\alpha'(\pi e_2) = \eta \cdot \alpha'(e_1)$, by Lemma 6.2.2. They are independent of the choice of the representatives.

It remains to show $\zeta \cdot \eta = \pi$. By the commutative diagram, we know that

$$\pi\alpha(e_2) = c' \circ c((\alpha/\pi)(\pi e_2)) = c'(\alpha'(\pi e_2)) = \eta c'(\alpha'(e_1)) = \eta\alpha(e_1) = \zeta\eta\alpha(e_2),$$

and hence $\zeta \cdot \eta = \pi$. □

Remark 6.2.5. *For any $\phi \in im(\iota)$, we could pick a representative of $\iota^{-1}(\phi)$ as following.*

$$\begin{array}{ccccc} \pi M & \hookrightarrow & M' & \hookrightarrow & M \\ \downarrow{\scriptstyle \alpha/\pi} & & \downarrow{\scriptstyle \alpha'} & & \downarrow{\scriptstyle \alpha} \\ R & \xrightarrow{\phi(\eta)} & R & \xrightarrow{\phi(\zeta)} & R \end{array}$$

In the above diagram, α sends e_2 (resp. e_1) to 1 (resp. $\phi(\zeta)$), and α' sends e_1 (resp. πe_2) to 1 (resp. $\phi(\eta)$). By an abuse of notation, we no longer distinguish between $\phi(\zeta)$ and ζ; similarly for η.

We know that

$$\begin{aligned} \mathrm{Hom}_{\mathcal{O}}(\mathcal{O}[\zeta, \eta]/(\zeta\eta - \pi), R) &= \mathrm{Hom}_{\mathcal{O}}(\mathcal{O}\langle\zeta, \eta\rangle/(\zeta\eta - \pi), R) \\ &= \mathrm{Hom}_{\mathcal{O}}(\mathrm{Spf}\, R, \mathrm{Spf}\, \mathcal{O}\langle\zeta, \eta\rangle/(\zeta\eta - \pi)). \end{aligned}$$

Proposition 6.2.6. *The functor $F_{[s,s']}$ is represented by the formal scheme $\widehat{\Omega}_{[s,s']}$.*

Proof. By Corollary 6.2.4, we identify $F_{[s,s']}$ with a subfunctor of the formal scheme $\mathrm{Spf}(\mathcal{O}\langle\zeta,\eta\rangle/(\zeta\eta-\pi))$. As usual we identify $\mathrm{Spf}(\mathcal{O}\{\zeta,\eta\}/(\zeta\eta-\pi))$ with the open subscheme of $\widehat{\mathbb{P}}_{[s,s']}$ which is the complement of the points at infinity $\bar{\zeta}=\infty$ and $\bar{\eta}=\infty$ of the two components of the special fiber.

By Remark 6.2.5, the condition $(**)$ can be expressed in terms of the images $\bar{\zeta}$ and $\bar{\eta}$ of ζ and η in $R/\pi R$: for all $a\in k-\{0\}$, $\bar{\zeta}-a$ and $\bar{\eta}-a$ are invertible in $R/\pi R$. Thus, $F_{[s,s']}$ is the subfunctor of $\widehat{\mathbb{P}}_{[s,s']}-(\{\bar{\zeta}=\infty\}\cup\{\bar{\eta}=\infty\})$ which represents the open which is the complement of the k-rational points of the two components of the special fiber $\mathrm{Spec}(k[\bar{\zeta},\bar{\eta}]/(\bar{\zeta}\bar{\eta}))$ with the exception of the singular point $\bar{\zeta}=\bar{\eta}=0$, which is to say $\widehat{\Omega}_{[s,s']}$. □

Remark 6.2.7. *We give another more direct proof. By Lemma 6.1.2,* $\zeta-a_i,\eta-a_i\in R^*$ *for* $1\le i\le q-1$. *Therefore,*

$$\begin{aligned}
F_s(R)&=\{(\zeta,\eta)\in R^2\mid(\zeta-a_i),(\eta-a_i)\in R^*,1\le i\le q-1 \text{ and } \zeta\eta=\pi\}\\
&=Hom_{\mathcal{O}}(\mathcal{O}[\zeta,\eta][(\zeta-a_i)^{-1},(\eta-a_i)^{-1}\mid 1\le i\le q-1]/(\zeta\eta-\pi),R)\\
&=Hom_{\mathcal{O}}(\mathcal{O}\langle\zeta,\eta\rangle\langle(\zeta-a_i)^{-1},(\eta-a_i)^{-1}\mid 1\le i\le q-1\rangle/(\zeta\eta-\pi),R)\\
&=Hom_{\mathcal{O}}(\mathrm{Spf}\,R,\widehat{\Omega}_{[s,s']})\\
&=\widehat{\Omega}_{[s,s']}(R).
\end{aligned}$$

6.3 The functorial explanation for the open immersion $\widehat{\Omega}_s\hookrightarrow\widehat{\Omega}_{[s,s']}$

The open immersion $\widehat{\Omega}_s\hookrightarrow\widehat{\Omega}_{[s,s']}$ is, with the identifications made above, the restriction of the open immersion $\mathrm{Spf}(\mathcal{O}\langle\zeta,\zeta^{-1}\rangle)\hookrightarrow\mathrm{Spf}(\mathcal{O}\langle\zeta,\eta\rangle/(\zeta\eta-\pi))$. The morphism $F_s\to F_{[s,s']}$ defined by functorially associating to each map $\alpha:M\to\mathcal{L}$ the commutative diagram:

$$\begin{array}{ccccc}
\pi M & \hookrightarrow & M' & \hookrightarrow & M\\
\downarrow{\scriptstyle \alpha/\pi} & & \downarrow{\scriptstyle \alpha|_{M'}} & & \downarrow{\scriptstyle \alpha}\\
\mathcal{L} & \xrightarrow{\pi} & \mathcal{L} & \xrightarrow{\mathrm{id}} & \mathcal{L}
\end{array}$$

identifies F_s with the subfunctor of $F_{[s,s']}$ consisting of the isomorphism class of diagrams where c' is invertible.

Similarly the open immersion $\widehat{\Omega}_{s'}\hookrightarrow\widehat{\Omega}_{[s,s']}$ is the restriction of the immersion $\mathrm{Spf}(\mathcal{O}\{\eta,\eta^{-1}\})\hookrightarrow\mathrm{Spf}(\mathcal{O}\{\zeta,\eta\}/(\zeta\eta-\pi))$. By functorially associating to each map $\alpha':M'\to\mathcal{L}'$ the commutative diagram:

$$\begin{array}{ccccc}
\pi M & \hookrightarrow & M' & \hookrightarrow & M\\
\downarrow{\scriptstyle \tilde{\alpha'}/\pi} & & \downarrow{\scriptstyle \alpha'} & & \downarrow{\scriptstyle \tilde{\alpha'}}\\
\mathcal{L}' & \xrightarrow{\mathrm{id}} & \mathcal{L}' & \xrightarrow{\pi} & \mathcal{L}'
\end{array}$$

identifies $F_{s'}$ with the subfunctor of $F_{[s,s']}$ consisting of the isomorphism class of diagrams where c is invertible. And $\tilde{\alpha'}$ is the natural extension of α' to M.

Besides, it is not hard to translate the canonical isomorphism between $\widehat{\Omega}_{[s,s']}$ and $\widehat{\Omega}_{[s',s]}$ to the natural isomorphism between $F_{[s,s']}$ and $F_{[s,s']}$. The following diagram is insightful.

$$\begin{array}{ccccccc}
\pi M' & \hookrightarrow & \pi M & \hookrightarrow & M' & \hookrightarrow & M\\
\downarrow{\scriptstyle \alpha'/\pi} & & \downarrow{\scriptstyle \alpha/\pi} & & \downarrow{\scriptstyle \alpha'} & & \downarrow{\scriptstyle \alpha}\\
\mathcal{L}' & \xrightarrow{c'} & \mathcal{L} & \xrightarrow{c} & \mathcal{L}' & \xrightarrow{c'} & \mathcal{L}
\end{array}$$

7 The functor of Drinfeld

From the functors F_s and $F_{[s,s']}$ defined above, we obtain a "modular" description of the open affines $\widehat{\Omega}_s$ and $\widehat{\Omega}_{[s,s']}$, which make up the formal scheme $\widehat{\Omega}$. A variant due to Drinfeld, which we will now describe, allows us to describe $\widehat{\Omega}$ directly in terms of a unique functor F defined on the category **Nilp** of $\mathcal{O}$-algebras where π is nilpotent. It is clear that **Nilp** is a subcategory of **Compl**. Thus, we may naturally regard the collection of functors $\{F_s\}_s$ and $\{F_{[s,s']}\}_{[s,s']}$ as being defined on **Nilp**.

If B is an $\mathcal{O}$-algebra, we write $B[\Pi]$ for the quotient of the algebra of polynomials $B[X]$ by the ideal generated by $X^2-\pi$: it's thus a free B-module of rank 2, generated by 1 element Π (the image of X), which satisfies $\Pi^2=\pi$. The algebra $B[\Pi]$ carries a $\mathbb{Z}/2\mathbb{Z}$-grading, such that the elements of B are of degree 0 and Π is of degree 1.

7.1 The functor F

Definition 7.1. *Let $B \in$ **Nilp** and $S = \operatorname{Spec} B$. Let $F(B)$ (or sometimes $F(S)$) denote the collection of isomorphism classes of quadruples (η, T, u, r) consisting of the following:*

(i) $\eta = \eta_0 \oplus \eta_1$ is a constructible sheaf of finitely generated flat $\mathbb{Z}/2\mathbb{Z}$-graded $\mathcal{O}[\Pi]$-modules, on S with the Zariski topology.

(ii) $T = T_0 \oplus T_1$ is a sheaf of $\mathbb{Z}/2\mathbb{Z}$-graded $\mathcal{O}_S[\Pi]$-modules such that the homogeneous components T_0 and T_1 are invertible sheaves on S.

(iii) u is an $\mathcal{O}[\Pi]$-homomorphism of degree 0 from η to T, such that $u \otimes_{\mathcal{O}} \mathcal{O}_S : \eta \otimes_{\mathcal{O}} \mathcal{O}_S \to T$ is surjective.

(iv) r is a K-isomorphism from the constant sheaf $\underline{K}_S^2$ to the sheaf $\eta_0 \otimes_{\mathcal{O}} K$.

These data are required to satisfy the following conditions:

[C1] Write $S_i \subset S$ for the zero locus of the morphism $\Pi : T_i \to T_{i+1}$ $(i = 0, 1)$; the restriction $\eta_i|_{S_i}$ is a constant sheaf with stalk isomorphic to $\mathcal{O}^2$.

[C2] For all geometric points x of S, write $T(x) = T \otimes_B k(x)$; the map $\eta_x/\Pi\eta_x \to T(x)/\Pi T(x)$ induced by u is injective.

[C3] For $i = 0, 1$, $(\wedge^2\eta_i)|_{S_i} = \pi^{-i}(\wedge^2(\Pi^i r\underline{\mathcal{O}}^2))|_{S_i}$.

Remark 7.1.1.

1 The definition of η means that there is a decomposition of S with disjoint locally closed subschemes where η is a locally constant sheaf associated to a flat $\mathbb{Z}/2\mathbb{Z}$-graded $\mathcal{O}[\Pi]$-module.

2 If f is a morphism from $\mathcal{F}_1$ to $\mathcal{F}_2$, where $\mathcal{F}_i$'s are $\mathcal{O}_X$-modules on X, a scheme. Then the zero locus of f is

$$\{x \mid f_x : \mathcal{F}_1 \otimes k(x) \to \mathcal{F}_2 \otimes k(x) \text{ is a zero map}\}.$$

In the above case, S_i is the collection of $x \in S$ such that $\Pi|_{T_i(x)}$ is the zero map.

3 It is clear that the above definition of $F(S)$ makes sense for S which are not necessarily affine, but also for all $\mathcal{O}$-schemes S such that the image of π is nilpotent (i.e. an $(\mathcal{O}/\pi^n\mathcal{O})$-scheme). We denote the category formed by these schemes by $\mathbf{Nilp}_{Sch}$.

Introducing Π is just a way to simplify the notation. We now give some explanations to clarify its essentiality in the above definition.

η is just a collection of two constructible sheaves of $\mathcal{O}$-modules with two morphisms $\Pi_0 : \eta_0 \to \eta_1$ and $\Pi_1 : \eta_1 \to \eta_0$ such that $\Pi_1 \circ \Pi_0 = \pi$ (on η_0) and $\Pi_0 \circ \Pi_1 = \pi$ (on η_1).

For T, it could be translated as the same as η.

The map $u : \eta \to T$ is just a pair of maps (u_0, u_1) where $u_i : \eta_i \to T_i$. The $\mathcal{O}[\Pi]$-linearity of u is equivalent to the commutativity of the following diagram (of period 2):

$$\begin{array}{ccccccccc}
\xrightarrow{\Pi_1} & \eta_0 & \xrightarrow{\Pi_0} & \eta_1 & \xrightarrow{\Pi_1} & \eta_0 & \xrightarrow{\Pi_0} \\
& \downarrow{\scriptstyle u_0} & & \downarrow{\scriptstyle u_1} & & \downarrow{\scriptstyle u_0} & \\
\xrightarrow{\Pi_1} & T_0 & \xrightarrow{\Pi_0} & T_1 & \xrightarrow{\Pi_1} & T_0 & \xrightarrow{\Pi_0}
\end{array}$$

And the surjectivity of $u \otimes \mathcal{O}_S$ is equivalent to the surjectivity of $u_i \otimes \mathcal{O}_S$ $(i = 1, 2)$.

The injectivity condition [C2] can be reformalized using these components as the injectivity of the following two maps:

$$(\eta_0/\Pi_1(\eta_1))_x \longrightarrow (T_0/\Pi_1(T_1)) \otimes_{\mathcal{O}} k(x),$$
$$(\eta_1/\Pi_0(\eta_0))_x \longrightarrow (T_1/\Pi_0(T_0)) \otimes_{\mathcal{O}} k(x).$$

Now we shall explain why F becomes a functor, and more precisely it is actually a "Zariski sheaf".

Lemma 7.1.2. *If $\Phi : X \to Y$ is a morphism of schemes, then the inverse image of a constant sheaf (resp. a constructible sheaf) is still a constant sheaf (resp. a constructible sheaf).*

Proof. For constant sheaves, it is clear.

For a constructible sheaf η on Y, we may assume $Y = \cup_i Y_i$ where Y_i's are locally closed subschemes of Y such that $\eta|_{Y_i}$ is a constant sheaf. Then $X_i := \Phi^{-1}(Y_i)$ is a locally closed subscheme of X for each i. Then,

$$(\Phi^{-1}\eta)|_{X_i} = \Phi^{-1}(\eta|_{Y_i})$$

is a constant sheaf. Therefore, $\Phi^{-1}\eta$ is still a constructible sheaf. □

For a morphism $\Phi : S \to S'$ in $\mathbf{Nilp}_{Sch}$, we will give a map from $F(S')$ to $F(S')$ which is functorial in S and S'.

For any $(\eta', T', u', r') \in F(S')$, by Lemma 7.1.2 and the fact that taking the inverse image preserves invertible sheaves, (η, T, u, r) is an element of $F(S)$ where $\eta = \Phi^{-1}\eta'$, $T = \Phi^* T'$, $u = \Phi^{-1}u'$ and $r = \Phi^{-1}r'$. The conditions in Definition 7.1 would be satisfied since these condition are local and taking the inverse image preserves stalks.

Moreover, we will show that F is a "Zariski" sheaf. In other words, F admits the following property.

Proposition 7.1.3. *If $S \in \mathbf{Nilp}_{Sch}$ is a scheme admitting an open covering $S = \cup_i S_i$. Then we have the following sequence exact.*

$$F(S) \longrightarrow \prod_i F(S_i) \rightrightarrows \prod_{i,j} F(S_i \cap S_j) \tag{1}$$

Proof. It is clear for the theorems of gluing sheaves and gluing morphisms of sheaves.

Note that for any open subscheme $U \subseteq S$, we have:

$$F(S) \longrightarrow F(U)$$
$$(\eta, T, u, r) \longmapsto (\eta|_U, T|_U, u|_U, r|_U).$$

□

7.2 The goal and the method

Our goal is to prove the following proposition.

Proposition 7.2.1. *The functor F is represented by the formal scheme $\widehat{\Omega}$.*

For this, we will define for each vertex s (resp. for each edge $[s,s']$) a morphism of functors $\iota_s : F_s \to F$ (resp. $\iota_{[s,s']} : F_{[s,s']} \to F$), in a manner compatible with patching, which is to say, compatible with the open immersions $F_s \hookrightarrow F_{[s,s']}$ and $F_{s'} \hookrightarrow F_{[s,s']}$. In this way, we define a morphism of functors, $\iota : \widehat{\Omega} \to F$, and we will show later that it is an isomorphism.

We now explain why we could define a morphism of functors $\widehat{\Omega} \to F$.

For any $B \in \mathbf{Nilp}$ and any $\phi \in \widehat{\Omega}(B)$, we will construct an associated element in $F(B)$.

Since $S = \operatorname{Spf} B$ is quasi-compact and $\widehat{\Omega} = \bigcup_T \widehat{\Omega}_T$ is a cofinal open covering, there is a finite tree T such that ϕ factors through $\widehat{\Omega}_T \hookrightarrow \widehat{\Omega}$. Then since $\widehat{\Omega}_T$, $\widehat{\Omega}_{[s,s']}$ and $\widehat{\Omega}_s$ are all affine, we can verify that

$$S_{[s,s']} := S \times_{\hat{\Omega}} \hat{\Omega}_{[s,s']} = S \times_{\hat{\Omega}_T} \hat{\Omega}_{[s,s']},$$
$$S_s := S \times_{\hat{\Omega}} \hat{\Omega}_s = S \times_{\hat{\Omega}_T} \hat{\Omega}_s$$

are both affine.

By gluing morphisms of formal schemes, we can verify that $\widehat{\Omega}$ is a "Zariski" sheaf and hence

$$\widehat{\Omega}(S) \longrightarrow \prod_{[s,s'],\text{edges}} \widehat{\Omega}(S_{[s,s']}) \rightrightarrows \prod_{s,\text{vertexes}} \widehat{\Omega}(S_s)$$

is exact.

Write

$$\phi \mapsto (\phi_{[s,s']})_{[s,s']} \mapsto (\phi_s)_s.$$

By definitions of $S_{[s,s']}$ and S_s, we know that $\phi_{[s,s']} \in F(S_{[s,s']})$ and $\phi_s \in F(S_s)$. By the exactness of (1), we could obtain a unique element in $F(S)$ by $(\iota_{[s,s']}(\phi_{[s,s']}))_{[s,s']} \mapsto (\iota_s(\phi_s))_s$. We send ϕ to this element. Thus, we obtain the morphism from $\widehat{\Omega}$ to F.

7.3 The definition of $F_s \to F$

We begin by remarking that each vertex s of the tree is represented by a lattice $M \subset K^2$ such that $\bigwedge^2 M = \bigwedge^2(\mathcal{O}^2)$, or $\bigwedge^2 M = \pi^{-1}\bigwedge^2(\mathcal{O}^2)$; such a representative M is unique, and the two possibilities are mutually exclusive: they define a partition on the collection of vertices, into even vertices (admitting a representative M such that $\bigwedge^2 M = \bigwedge^2(\mathcal{O}^2)$) and odd vertices (admitting a representative M such that $\bigwedge^2 M = \pi^{-1}\bigwedge^2(\mathcal{O}^2)$).

Note that a neighbouring vertex of an even vertex (resp. odd) is odd (resp. even). We suppose in what follows that the representatives M of vertices are chosen such that $\bigwedge^2 M = \bigwedge^2(\mathcal{O}^2)$ or $\bigwedge^2 M = \pi^{-1}\bigwedge^2(\mathcal{O}^2)$.

Similarly, we always orient the edges $[s,s']$ such that s is odd and s' is even. Thus, we have representatives M and M' satisfying: $\bigwedge^2 M = \pi^{-1}\bigwedge^2(\mathcal{O}^2)$, $\bigwedge^2 M' = \bigwedge^2(\mathcal{O}^2)$ and $\pi M \subset M' \subset M$.

In the following, we will define $F_s \to F$. We distinguish two cases, according to whether s is even or odd.

Definition of $F_s \to F$ for $s = [M]$ odd $[\bigwedge^2 M = \pi^{-1}\bigwedge^2(\mathcal{O}^2)]$. Giving a point in $F_s(B)$ is equivalent to giving an invertible sheaf $\mathcal{L}$ (which is actually free of rank 1) on $S = \operatorname{Spec} B$ and an $\mathcal{O}$-morphism $\alpha : \underline{M}_S \to \mathcal{L}$, such that the map $\alpha(x) : \underline{M}_S/\pi\underline{M}_S \to \mathcal{L} \otimes_B k(x)$ is injective for any geometric point x of S.

To such a point corresponds the point of $F(B)$ defined by the following diagram:

$$\begin{array}{ccccc}
\eta_0 = \underline{M} & \xrightarrow{\Pi=\mathrm{id}} & \eta_1 = \underline{M} & \xrightarrow{\Pi=\pi} & \eta_0 = \underline{M} \\
{\scriptstyle u_0=\alpha}\downarrow & & {\scriptstyle u_1=\alpha}\downarrow & & {\scriptstyle u_0=\alpha}\downarrow \\
T_0 = \mathcal{L} & \xrightarrow{\Pi=\mathrm{id}} & T_1 = \mathcal{L} & \xrightarrow{\Pi=\pi} & T_0 = \mathcal{L}
\end{array}$$

and by the isomorphism $r : \underline{K^2} \xrightarrow{\sim} \underline{M} \otimes_{\mathcal{O}} K$ which corresponds to the inclusion $M \hookrightarrow K^2$. It is easy to see that it indeed defines a quadruple. Then we shall check if the conditions of Definition 7.1 are satisfied.

It is clear that $S_0 = \varnothing$ and $S_1 = S$. Since η is a constant sheaf, the condition [C1] holds naturally. The condition [C2] holds for the definition of F_s. For the condition [C3], it suffices to show $(\wedge^2\eta_1) = \pi^{-1}(\wedge^2(\Pi r\underline{\mathcal{O}^2}))$. Since $\Pi|_{\eta_0} = \mathrm{id}$ and r is defined via the inclusion $M \hookrightarrow K^2$, it suffices to show $(\wedge^2\underline{M}) = \pi^{-1}(\wedge^2(\underline{\mathcal{O}^2}))$. This is true because s is odd.

Definition of $F_{s'} \to F$ for $s' = [M']$ even [$\bigwedge^2 M' = \bigwedge^2(\mathcal{O}^2)$]. To a point of $F_{s'}(B)$, represented by $\alpha' : M' \to \mathcal{L}'$, we associate the point of $F(B)$ defined by the diagram:

$$\begin{array}{ccccc} \eta_0 = M' & \xrightarrow{\Pi=\pi} & \eta_1 = M' & \xrightarrow{\Pi=\mathrm{id}} & \eta_0 = M' \\ {\scriptstyle u_0=\alpha'}\downarrow & & {\scriptstyle u_1=\alpha'}\downarrow & & {\scriptstyle u_0=\alpha'}\downarrow \\ T_0 = \mathcal{L}' & \xrightarrow{\Pi=\pi} & T_1 = \mathcal{L}' & \xrightarrow{\Pi=\mathrm{id}} & T_0 = \mathcal{L}' \end{array}$$

and by the isomorphism $r : \underline{K^2} \xrightarrow{\sim} \underline{M'} \otimes K$ which corresponds with the inclusion $M' \hookrightarrow K^2$. It is easy to show it satisfies all the conditions in Definition 7.1 by the similar argument.

7.4 The definition of $F_{[s,s']} \to F$

It remains to define the maps $F_{[s,s']} \to F$ for an edge $[s, s']$. Suppose that the orientation and the representatives M and M' are chosen as in the previous subsection. Giving a point of $F_{[s,s']}(B)$ corresponds with giving an isomorphism class of commutative diagrams satisfying condition $(**)$ of Definition 6.3:

$$\begin{array}{ccccc} \pi M & \hookrightarrow & M' & \hookrightarrow & M \\ \downarrow{\scriptstyle \alpha/\pi} & & \downarrow{\scriptstyle \alpha'} & & \downarrow{\scriptstyle \alpha} \\ \mathcal{L} & \xrightarrow{c} & \mathcal{L}' & \xrightarrow{c'} & \mathcal{L} \end{array}$$

and we denote this point as Φ.

By an abuse of notation, we could view $\mathcal{L}$ and $\mathcal{L}'$ as invertible sheaves on S. And we know that they are actually free of rank one.

We could see that $S = \operatorname{Spec} B$ is the union of two closed subschemes S_0 and S_1, where S_0 (resp. S_1) is the locus of points where c (resp. c') vanishes. Write $\mathcal{U} \subset S_1$ (resp. $\mathcal{U}' \subset S_0$) for the open where c' (resp. c) is invertible. Note that $\mathcal{U}$ and $\mathcal{U}'$ are principal open subsets and hence affine.

The open immersion of $\mathcal{U} \to S$ (resp. $\mathcal{U}' \to S$) defines a map j (resp. j') from $F_{[s,s']}(S)$ to $F_{[s,s']}(\mathcal{U})$ (resp. $F_{[s,s']}(\mathcal{U})$). Similarly, the closed immersion $S_0 \cap S_1 \to S$ defines a map j''.

$j(\Phi)$ comes from the point of $F_s(\mathcal{U})$ defined by $\alpha : M \to \mathcal{L} \otimes_B \mathcal{O}_S(\mathcal{U})$. The construction in the previous subsection yields the following point of $F(\mathcal{U})$:

$$\begin{array}{ccccc} \underline{M} & \xrightarrow{\mathrm{id}} & \underline{M} & \xrightarrow{\pi} & \underline{M} \\ \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \alpha} \\ \mathcal{L}|_{\mathcal{U}} & \xrightarrow[\mathrm{id}]{} & \mathcal{L}|_{\mathcal{U}} & \xrightarrow[\pi]{} & \mathcal{L}|_{\mathcal{U}} \end{array}$$

and where r is defined by the inclusion $M \hookrightarrow K^2$.

Or, what amounts to the same thing by using the isomorphism $c' : \mathcal{L}' \xrightarrow{\sim} \mathcal{L}$, the point is defined by the diagram:

$$\begin{array}{ccccc} \underline{M} & \xrightarrow{\mathrm{id}} & \underline{M} & \xrightarrow{\pi} & \underline{M} \\ \downarrow{\scriptstyle (c')^{-1}\alpha} & & \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle (c')^{-1}\alpha} \\ \mathcal{L}'|_{\mathcal{U}} & \xrightarrow[c']{\sim} & \mathcal{L}|_{\mathcal{U}} & \xrightarrow[c]{} & \mathcal{L}'|_{\mathcal{U}} \end{array} \qquad (*)$$

Note that $(c')^{-1}\alpha$ is an extension to M of the arrow $\alpha' : M' \to \mathcal{L}'$ (it is not defined above $\mathcal{U}$). The two definitions coincide since the two diagrams are isomorphic.

Similarly, $j'(\Phi)$ corresponds to the point of $F_{s'}(\mathcal{U}')$ defined by $\alpha' : M' \to \mathcal{L}'$. It is associated with the point of $F(\mathcal{U}')$ defined by:

$$\begin{array}{ccccc} \underline{M'} & \xrightarrow{\text{id}} & \underline{M'} & \xrightarrow{\pi} & \underline{M'} \\ \downarrow{\scriptstyle \alpha'} & & \downarrow{\scriptstyle \alpha'} & & \downarrow{\scriptstyle \alpha'} \\ \mathcal{L}'|_{\mathcal{U}'} & \xrightarrow[\text{id}]{} & \mathcal{L}'|_{\mathcal{U}'} & \xrightarrow[\pi]{} & \mathcal{L}'|_{\mathcal{U}'} \end{array}$$

(with r defined by the inclusion $M' \hookrightarrow K^2$). Applying the isomorphism $c : \mathcal{L} \xrightarrow{\sim} \mathcal{L}'$, we obtain a diagram isomorphic to the one above again.

$$\begin{array}{ccccc} \underline{M'} & \xrightarrow{\pi} & \underline{M'} & \xrightarrow{\text{id}} & \underline{M'} \\ \downarrow{\scriptstyle \alpha'} & & \downarrow{\scriptstyle (c')^{-1}\alpha'} & & \downarrow{\scriptstyle \alpha'} \\ \mathcal{L}'|_{\mathcal{U}'} & \xrightarrow[c']{} & \mathcal{L}|_{\mathcal{U}'} & \xrightarrow[c]{\sim} & \mathcal{L}'|_{\mathcal{U}'} \end{array} \qquad (**)$$

where $c^{-1}\alpha'$ extends the arrow $\alpha/\pi : \pi M \to \mathcal{L}$ to M'.

Finally, the complement of the union $\mathcal{U} \cup \mathcal{U}'$ is equal to the intersection $J = S_0 \cap S_1$, and J is also affine. Then $j''(\Phi)$ corresponds to the point of $F_{s'}(J)$ via the diagram:

$$\begin{array}{ccccc} \underline{M'} & \hookrightarrow & \underline{M} & \xrightarrow{\pi} & \underline{M'} \\ \downarrow{\scriptstyle \alpha'} & & \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \alpha'} \\ \mathcal{L}'|_J & \xrightarrow[c']{} & \mathcal{L}|_J & \xrightarrow[c]{} & \mathcal{L}'|_J \end{array} \qquad (***)$$

with r corresponding again with the inclusion $M' \hookrightarrow K^2$.

We thus have three points of F with values respectively in the schemes $\mathcal{U}, \mathcal{U}'$ and J, which are described by the three diagrams $(*)$, $(**)$ and $(***)$ above. We must now explain how they glue to yield a single point (η, T, u, r) of $F(S)$.

The most simple part is to define T: we take $T_0 = \mathcal{L}'$ and $T_1 = \mathcal{L}$ on the all of S, where the morphism Π is given by c' and c.

Next, we shall give the definition of the sheaf η.

Consider a sheaf, $\mathcal{F}$, on $S_0 \bigsqcup S_1$ which is constant with value M on S_1 and of value M' on S_0. Let ϕ be the natural morphism from $S_0 \bigsqcup S_1$ to S.

Lemma 7.4.1. *$(\phi_*\mathcal{F})|_J$ is a constant sheaf of value $M \oplus M'$, where $(\phi_*\mathcal{F})|_J$ is the inverse image of $(\phi_*\mathcal{F})$ corresponding to $J \to S$.*

Proof. Let V be any open of J. If we set

$$(\phi_*\mathcal{F})|'_J(V) = \varinjlim_{V \subset U, U \text{ open in } S} \phi_*\mathcal{F}(U) \ ,$$

then $(\phi_*\mathcal{F})|_J$ is the shifification of $(\phi_*\mathcal{F})|'_J$.

For any open U in S such that $V \subset U$ which implies that $U \cap J \neq \varnothing$, we have

$$\phi_*\mathcal{F}(U) = \mathcal{F}(U \cap S_0) \oplus \mathcal{F}(U \cap S_1) = M \oplus M'.$$

Thus, $(\phi_*\mathcal{F})|'_J$ is already a constant sheaf of value $M \oplus M'$, then so does $(\phi_*\mathcal{F})|_J$. □

Remark 7.4.2. *For any open U in S, we thus have a natural map from $\phi_*\mathcal{F}(U)$ to $(\phi_*\mathcal{F})|_J(U\cap J)$ and we denote the image of f $(\in \phi_*\mathcal{F}(U))$ by $f|_{U\cap J}$.*

Let ι (resp. ι') be the morphism from M (resp. M') to $M\oplus M'$ sending m (resp. m) to $(m,\pi m)$ (resp. (m',m')).

We define the sheaf η_0 (resp. η_1) as the subsheaf of sections of $\phi_*(\mathcal{F})$ which, above J, take values in the submodule $\iota'(M')$ (resp. $\iota(M)$).

To be explicit, we give the expression of η_0: for any open U in S,

$$\begin{aligned}\eta_0(U) &= \{f\in\phi_*\mathcal{F}(U)\mid f|_{U\cap J}\in\iota'(M')\subset M\oplus M'\}\\ &= \begin{cases}\phi_*\mathcal{F}(U)=\underline{M}'(U\cap S_0)\oplus\underline{M}(U\cap S_1) & \text{if } U\cap J=\varnothing\\ \iota'(M')\subset M\oplus M' & \text{if } U\cap J\neq\varnothing\end{cases},\end{aligned}$$

and the expression of η_1 is similar.

The morphisms $\Pi:\eta_0\to\eta_1$ and $\eta_1\to\eta_0$ are obtained by restriction from the endomorphisms of $\phi_*\mathcal{F}$ given respectively by:

$$\begin{cases} M\xrightarrow{id}M\\ M'\xrightarrow{\pi}M'\end{cases}\quad\text{and}\quad\begin{cases} M\xrightarrow{\pi}M\\ M'\xrightarrow{id}M'\end{cases}$$

We denote the two endomorphisms as φ_0 and φ_1. And let $\Pi|_{\eta_i}=\varphi_i|_{\eta_i}$. Their relations are shown in the following diagram.

$$\begin{array}{ccc}\phi_*\mathcal{F} & \xrightarrow{\varphi_i} & \phi_*\mathcal{F}\\ \uparrow & & \uparrow\\ \eta_i & \xrightarrow{\Pi} & \eta_{i+1}\end{array}$$

It is clearly that η_0 and η_1 are constructable sheaves over S. More precisely, when we restrict η_i on S_0, S_1 or J, they are constant sheaves.

Alternatively, the definition of η is summarized by the following diagram:

$$\eta_0\xrightarrow{\Pi}\eta_1\xrightarrow{\Pi}\eta_0$$

$$\begin{array}{lccccc}\text{restrict to }\mathcal{U} & M & \xrightarrow{\mathrm{id}} & M & \xhookrightarrow{\pi} & M\\ S_1\wr & \uparrow & & \| & & \uparrow\\ \text{restrict to } J & M' & \hookrightarrow & M & \xhookrightarrow{\pi} & M'\\ S_0\wr & \| & & \downarrow\pi & & \|\\ \text{restrict to }\mathcal{U}' & M' & \xhookrightarrow{\pi} & M' & \xrightarrow{\mathrm{id}} & M'\end{array}$$

Note in particular that $\eta_0|_{S_0}$ is constant of value M' and that $\eta_1|_{S_1}$ is constant of value M.

The isomorphism $r:K^2\xrightarrow{\sim}\eta_0\otimes_O K$ is defined by the (compatible) inclusions of M and M' into K^2.

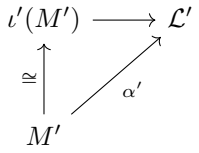
Finally, the morphism u_0 is given by α' on $S_0=\mathcal{U}'\cup J$ and by $(c')^{-1}\alpha$ on $\mathcal{U}$. Similarly, u_1 is equal to α on $S_1=\mathcal{U}\cup J$ and to $c^{-1}\alpha'$ on $\mathcal{U}'$.

We could give explicit expressions of u_0, u_1 and r on each open subset in S, since we have described η sufficiently explicit. For example, for any open $U\subseteq S$, we describe u_0 as follows:

- If $U\cap J=\varnothing$, then u_0 is given by the morphisms:

$$M\xrightarrow{(c')^{-1}\alpha}\mathcal{L}'\quad\text{and}\quad M'\xrightarrow{\alpha'}\mathcal{L}'.$$

- If $U \cap J \neq \varnothing$, then u_0 is determined by the following commutative diagram:



It is clear that we have thus constructed a point (η, T, u, r) of $F(S)$, and have defined the morphism $F_{[s,s']} \to F$ that we were after. Moreover, our construction shows that these morphisms glue the morphisms $F_s \to F$ and $F_{s'} \to F$ defined in the precious subsection.

Therefore, we obtain a morphism of functors, $\iota : \hat{\Omega} \to F$.

7.5 Some results following the definition

In this subsection, we will give some results obtained immediately from Definition 7.1.

Lemma 7.5.1. *S_i is a closed subscheme of S.*

Proof. Check locally. We may assume T_0 and T_1 are two constant sheaves $\mathcal{O}_{S'}$ on S', an open subscheme of S. Then Π is just multiplying f for some f in $\Gamma(\mathcal{O}_{S'}, S')$. Thus, the zero locus is just $V(f)$ and hence closed. □

Lemma 7.5.2. *$S = S_0 \cup S_1$.*

Proof. Since $\Pi^2 = \pi$, it is clear that the composition of $\Pi|_{T_0(x)}$ and $\Pi|_{T_1(x)}$ is a zero map. Since $T_i(x)$'s are k-linear spaces, one of $\Pi|_{T_i(x)}$'s must be zero, i.e. $x \in S_0$ or S_1. Thus, the assersion holds. □

Lemma 7.5.3. *The homogeneous components η_0 and η_1 are flat sheaves of $\mathcal{O}$-modules such that each stalk is free of rank 2. The action of Π defines injective maps*

$$\cdots \eta_0 \xrightarrow{\Pi} \eta_1 \xrightarrow{\Pi} \eta_0 \cdots,$$

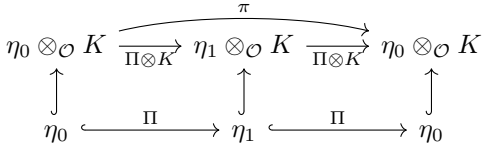
which satisfy the relation $\Pi^2 = \pi$.

Proof. First, we will show that η_0 and η_1 are flat $\mathcal{O}$-modules. Since $\mathcal{O}$ is a PID, it suffices to show they are sheaves of torsion-free $\mathcal{O}$-modules. By the definition of η, it suffices to show: if $M = M_0 \oplus M_1$ is a $\mathbb{Z}/2\mathbb{Z}$-graded $\mathcal{O}[\Pi]$-module, then M is $\mathcal{O}$-torsion-free. It is true for $\mathcal{O}$ is a PID and $\mathcal{O}[\Pi]$ is a free $\mathcal{O}$-module.

Then, we will show that each stalk of η_0 is a free $\mathcal{O}$-module of rank 2. Let x be a geometric point on S. Since $(\eta_i)_x$ $(i = 0, 1)$ is a finitely generated $\mathcal{O}$-module and torsion-free, it should be a free $\mathcal{O}$-module by the structure theorem for the finitely generated modules over a PID. Then $(\eta_0)_x$ is clearly free of rank 2 by the existence of r.

Finally, we will show that filtration consists of injective maps, which forces the rank of the stalk of η_1 to be the same as η_0, 2. Since $\Pi \circ \Pi = \pi$ and $(\eta_i)_x \xrightarrow{\pi} (\eta_i)_x$ is clearly injective for $(\eta_i)_x$ is torsion-free. Then Π must be injective, too. □

Besides, we have the following commutative diagram (Δ).



Thus, we have

$$\eta_0 \subseteq (\Pi \otimes K)^{-1}(\eta_1) = \pi^{-1}(\Pi(\eta_1)) \subseteq \eta_0 \otimes_{\mathcal{O}} K.$$

Definition 7.2. *Using r, we could define sub-sheaves N_0 and N_1 of the constant sheaf $\underline{K}_S^2$:*

$$N_0 = r^{-1}\eta_0$$

$$N_1 = r^{-1}(\Pi \otimes K)^{-1}(\eta_1) = \pi^{-1}r^{-1}(\Pi(\eta_1))$$

where we view η_i $(i = 1, 2)$ as a subsheaf of $\eta_i \otimes_{\mathcal{O}} K$.

These are sub-sheaves of $\mathcal{O}$-modules of maximal rank (the "edges") which are isomorphic respectively to η_0 and η_1. At each geometric point x of S, we have the inclusions:

$$N_{0,x} \subset N_{1,x} \subset \pi^{-1}N_{0,x} \subset K^2,$$

and thus a simple (vertex and edge) of the tree.

More precisely, via the conditions in the definition, we could get the following results.

Lemma 7.5.4. *Let x be a geometric point in S. The relationship between the lattices $N_{0,x}$ and $N_{1,x}$ is determined by the following cases:*

1. *If $\Pi|_{T_0(x)}$ is invertible, then $N_{0,x} = N_{1,x}$. Moreover, we have:*
$$\bigwedge^2 N_{1,x} = \pi^{-1} \bigwedge^2 (\mathcal{O}^2).$$
2. *If $\Pi|_{T_1(x)}$ is invertible, then $N_{1,x} = \pi^{-1}N_{0,x}$. Moreover, we have:*
$$\bigwedge^2 N_{0,x} = \bigwedge^2 (\mathcal{O}^2).$$
3. *If both $\Pi|_{T_0(x)} = 0$ and $\Pi|_{T_1(x)} = 0$, then we obtain a chain of strict inclusions $N_{0,x} \subsetneq N_{1,x} \subsetneq \pi^{-1}N_{0,x}$, with:*
$$\bigwedge^2 N_{0,x} = \bigwedge^2 (\mathcal{O}^2) \quad \text{and} \quad \bigwedge^2 N_{1,x} = \pi^{-1} \bigwedge^2 (\mathcal{O}^2).$$

Proof. Firstly, we shall show that if $x \in S_0$ (resp. $x \in S_1$) then

$$\bigwedge^2 N_{0,x} = \bigwedge^2 (\mathcal{O}^2) \quad (\text{ resp. } \bigwedge^2 N_{1,x} = \pi^{-1} \bigwedge^2 (\mathcal{O}^2) \text{ }).$$

By condition [C3],

$$\bigwedge^2 N_{0,x} = \bigwedge^2 r^{-1}\eta_{0,x} = r^{-1} \bigwedge^2 r(\mathcal{O}^2) = \bigwedge^2 \mathcal{O}^2 \text{ if } x \in S_0$$

$$\left(\text{resp} \bigwedge^2 N_{1,x} = \bigwedge^2 \pi^{-1}r^{-1}\Pi\eta_{1,x} = \pi^{-2}r^{-1}\Pi \bigwedge^2 \Pi r(\mathcal{O}^2) = \pi^{-1} \bigwedge^2 \mathcal{O}^2 \text{ if } x \in S_0 \right).$$

If $\Pi|_{T_0(x)}$ is invertible, then $T_1(x) = \Pi(T_0(x))$, which implies $T_1(x)/\Pi(T_0(x)) = 0$. By the condition [C2], we have $\eta_{1,x}/\Pi\eta_{0,x} = 0$, i.e. $\Pi|_{\eta_{0,x}}$ is surjective and hence bijective. Thus, $\eta_{0,x} = (\Pi \otimes K)^{-1}(\eta_{1,x})$ by the commutative diagram (Δ). Hence $N_{0,x} = N_{1,x}$. Besides, since $x \notin S_0$, then $x \in S_1$ by Lemma 7.5.2 Thus the assertion for $N_{1,x}$ holds.

The second assertion follows by the same arguement.

It remains to show that under the assumption of the third assertion, we indeed have a branch: $N_{0,x} \subsetneq N_{1,x} \subsetneq \pi^{-1}N_{0,x}$. Suppose $N_{0,x} = N_{1,x}$, then $\Pi(\eta_{0,x}) = \eta_{1,x}$. By the surjectivity of $u_i \otimes_{\mathcal{O}} \mathcal{O}_S$, we thus have $\Pi(T_0(x)) = T_1(x)$. Since $\Pi|_{T_i(x)}$'s are both zero, then $T_1(x) = 0$. It is impossible since T_1 is an invertible sheaf, a contradiction. The case where $\pi^{-1}N_{0,x} = N_{1,x}$ leads to contradictions for the same reason. □

Remark 7.5.5. *Since T_i's are invertible sheaves, then $T_i(x)$ are one-dimensional $k(x)$-vector spaces, and hence $\Pi|_{T_i(x)}$ is either invertible or zero.*

7.6 The functor represented by $\widehat{\Omega}$

We must still show that the morphism $\iota : \hat{\Omega} \to F$ is an isomorphism of functors: in other words, for an arbitrary point of $F(B)$ there is a unique corresponding point of $\widehat{\Omega}(B)$.

Fix a point $(\eta, T, u, r) \in F(B)$. We first deal with the case where T_i's are free (of rank 1) on $S = \operatorname{Spec} B$.

We associate to each edge $[s, s']$ of the tree, which we suppose is represented by two lattices M and M' such that $\pi M \subsetneq M' \subsetneq M$ with the conventions before, an subset $S_{[s,s']}$ of S defined in the following way: it's the collection of points x which satisfy, with the notations in Definition 7.2, the inclusions:

$$M' \subset N_{0,x} \quad \text{and} \quad M \subset N_{1,x}.$$

By Lemma 7.5.4, we deduce that this amounts to the following cases:

$$\begin{cases} \Pi|_{T_0(x)} \text{ is invertible:} & N_{0,x} = N_{1,x} = M, \\ \Pi|_{T_1(x)} \text{ is invertible:} & N_{0,x} = \pi N_{1,x} = M', \\ \text{neither is invertible:} & N_{0,x} = M',\ N_{1,x} = M. \end{cases}$$

Proposition 7.6.1. *We have $S_{[s,s']}$ is a Zariski open subscheme of S.*

Proof. It suffices to verify that the intersections $S_{[s,s']} \cap S_0$ and $S_{[s,s']} \cap S_1$ are Zariski opens, in S_0 and S_1 respectively by Lemmas 7.5.1 and 7.5.2.

The intersection $S_{[s,s']} \cap S_0$ is the locus of points of S_0 which satisfy:

$$N_{0,x} = M' \quad \text{and} \quad N_{1,x} = M \text{ or } \pi^{-1}M'.$$

As the sheaf η_0 is constant on S_0, N_0 is locally constant. The first condition above thus defines a subset $\mathcal{A} = \{x \in S_0 \mid N_{0,x} = M'\} \subset S_0$ which is both open and closed. On the open subset of $\mathcal{A}$ where $\Pi|_{T_1}$ is invertible, one has automatically $N_{1,x} = \pi^{-1}M'$. In other words, the complement of $S_{[s,s']} \cap S_0$ in $\mathcal{A}$ is contained in $S_0 \cap S_1$. This complement is defined in $S_0 \cap S_1$ by the condition $N_{1,x} \neq M$, and is thus closed (as well as open) since the sheaf N_1 is locally constant on S_1.

It follows that $S_0 \cap S_{[s,s']}$ is open in S_0; we could show similarly that $S_1 \cap S_{[s,s']}$ is open in S_1. Thus $S_{[s,s']}$ is open in S. □

By Lemma 7.5.4, we could see that $S_{[s,s']}$'s cover S. If $[s, s']$ and $[s, s'']$ are two distinct edges which intersect in s, then $S_{[s,s']} \cap S_{[s,s'']}$ is the open S_s, which is the locus of points satisfying $N_{0,x} = N_{1,x} = M$; note that S_s can also be defined in $S_{[s,s']}$ or $S_{[s,s'']}$ by the condition: $\Pi|T_0(x)$ is invertible. Moreover, for two edges $[s, s']$ and $[s'', s']$ with intersection s', the intersection $S_{[s,s']} \cap S_{[s'',s']}$ is the open $S_{s'}$ "made up of" the points x which satisfy $N_{0,x} = \pi N_{1,x} = M'$, and which is also defined in $S_{[s,s']}$ or $S_{[s'',s']}$ by the condition: $\Pi|T_1(x)$ is invertible.

The following diagram defines a point of the functor $F_{[s,s']}$ with values in $S_{[s,s']}$:

$$\begin{array}{ccccc}
\underline{\pi M} & \hookrightarrow & \underline{M'} & \hookrightarrow & \underline{M} \\
\downarrow & & \downarrow & & \downarrow \\
\pi N_1 & \longrightarrow & N_0 & \longrightarrow & N_1 \\
\downarrow{\scriptstyle \pi^{-1}\circ l} & & \downarrow{\scriptstyle r} & & \downarrow{\scriptstyle l} \\
\eta_1 & \xrightarrow{\Pi} & \eta_0 & \xrightarrow{\Pi} & \eta_1 \\
\downarrow{\scriptstyle u_1} & & \downarrow{\scriptstyle u_0} & & \downarrow{\scriptstyle u_1} \\
T_1 & \xrightarrow{\Pi} & T_0 & \xrightarrow{\Pi} & T_1
\end{array}$$

where $l = (\Pi \otimes K) \circ r$.

It is clear that the points thus obtained glue: on S_s for example, one obtains the points of the functor F_s defined by the composition

$$M = N_1 \xrightarrow[\cong]{r} \eta_1 \to T_1.$$

By gluing, we thus obtains a point of $\widehat{\Omega}(B)$, and it is easy to show that it maps to the initial point (η, T, u, r). It is easy to verify that this is the only point which ι maps to (η, T, u, r). Thus, this special case holds.

For separated $S =\in \mathbf{Nilp}_{Sch}$ and any $\Phi = (\eta, T, u, r) \in F(S)$, S admits an open affine covering $S = \cup_i S_i$ such that $T|_{S_i}$ is free. Then there is a point $\Phi_i \in \widehat{\Omega}(S_i)$ such that $\iota^{-1}(\Phi|_{S_i}) = \{\Phi_i\}$. Similarly, since S is separated, $S_i \cap S_j$ is affine, then we have $\Phi_{ij} \in \widehat{\Omega}(S_i \cap S_j)$ such that $\iota^{-1}(\Phi|_{S_i \cap S_j}) = \{\Phi_{ij}\}$. By gluing, we obtain a unique point in $\widehat{\Omega}(S)$ corresponding to Φ.

Finally, for general $S \in \mathbf{Nilp}_{Sch}$, write $S = \cup_l S_l$ where S_l's are affine open. Then S_l and $S_l \cap S_{l'}$ are separated. Therefore, Proposition 7.2.1 follows from the diagram below.

$$\begin{array}{ccccc}
\widehat{\Omega}(S) & \longrightarrow & \prod_i \widehat{\Omega}(S_i) & \rightrightarrows & \prod_{i,j} \widehat{\Omega}(S_i \cap S_j) \\
\downarrow{\scriptstyle \iota} & & \downarrow{\scriptstyle \cong} & & \downarrow{\scriptstyle \cong} \\
F(S) & \longrightarrow & \prod_l F(S_l) & \rightrightarrows & \prod_{l,l'} F(S_l \cap S_{l'})
\end{array}$$

8 The action of the group $\mathrm{PGL}_2(K)$

8.1 The action on the tree I,the set Ω, the rigid space Ω and the formal scheme $\widehat{\Omega}$

The group $\mathrm{GL}_2(K)$ acts naturally, through its quotient $\mathrm{PGL}_2(K)$, on the tree I: an element $g \in \mathrm{GL}_2(K)$ transforms the vertex $[M]$ (resp. the edge $[[M],[M']]$) to the vertex $[gM]$ (resp. to the edge $[[gM],[gM']]$).

One also has an action of the same group $\mathrm{PGL}_2(K)$ on the set $\Omega = \mathbb{P}^1(\mathbb{C}) - \mathbb{P}^1(K)$. It is clear that the map $\lambda : \Omega \to I_{\mathbb{R}}$ is equivariant for this action, and it thus follows that this group acts by automorphisms of the rigid analytic space Ω: the action permutes the different affinoid opens defined in §5. We see without difficulty that this action also yields an action on the formal scheme $\widehat{\Omega}$ by permuting the different formal affine opens.

More precisely, for any $g \in \mathrm{PGL}_2(K)$, we could give compatible morphisms (see the next subsection)

$$\widehat{\Omega}_s \xrightarrow{g} \widehat{\Omega}_{gs} \quad \text{and} \quad \widehat{\Omega}_{[s,s']} \xrightarrow{g} \widehat{\Omega}_{[gs,gs']}$$

to obtain an automorphism of $\widehat{\Omega}$, which gives an action on $\widehat{\Omega}$ by $\mathrm{PGL}_2(K)$. By taking the generic fibre, we obtain an action on Ω by $\mathrm{PGL}_2(K)$.

8.2 The action on the functors of Deligne and the functor of Drinfeld

The action above admits a description in terms of Deligne's functors: for $s = [M]$ and $gs = [gM]$, one has automorphisms of functors:

$$g : F_s \to F_{gs},$$

given by $g \cdot (\mathcal{L}, \alpha) = (\mathcal{L}, \beta)$ where β denotes the composition

$$gM \xrightarrow{g^{-1}} M \xrightarrow{\alpha} \mathcal{L}.$$

Moreover, for $s' = [M']$ such that $[s, s']$ is an edge, the morphism $g : F_{[s,s']} \to F_{[gs,gs']}$ is given by:

$$g \cdot (\mathcal{L}, \mathcal{L}', c, c', \alpha, \alpha') = (\mathcal{L}, \mathcal{L}', c, c', \alpha \circ g^{-1}, \alpha' \circ g^{-1}).$$

And it is easy to show this action is compatible with the open immersion $F_s \to F_{[s,s']}$.

By Yoneda embedding on **Compl**, we thus get unique morphisms

$$\widehat{\Omega}_s \xrightarrow{g} \widehat{\Omega}_{gs} \quad \text{and} \quad \widehat{\Omega}_{[s,s']} \xrightarrow{g} \widehat{\Omega}_{[gs,gs']},$$

which completes the discussion in the previous subsection.

It is a little more difficult to describe the action of $\mathrm{PGL}_2(K)$ on $\hat{\Omega}$ in terms of the functor F of Drinfeld. One such description is furnished by the following proposition.

Proposition 8.2.1. *The action of an element $g \in \mathrm{GL}_2(K)$ on the functor F is given by the following formula:*

$$g \cdot (\eta, T, u, r) = (\eta[n], T[n], u[n], \Pi^n \circ r \circ g^{-1}),$$

where n denotes the valuation of $\det g$, and $[n]$ the shift by $n \pmod 2$ of the grading of (η, T, u).

Proof. Note that the shift ensures that the normalization condition [C3] of Definition 7.1 is satisfied for the image. In fact, we write r_1 for the composition:

$$r_1 : K^2 \xrightarrow{g^{-1}} K^2 \xrightarrow{r} \eta_0 \otimes_{\mathcal{O}} K \xrightarrow{\Pi^n} \eta[n]_0 \otimes_{\mathcal{O}} K.$$

The zero locus of the morphism $\Pi : T[n]_i \to T[n]_{i+1}$ is $S_{i'}$, with $i' \equiv i+n \pmod 2$, above which one has:

$$\begin{aligned}\bigwedge^2 \eta[n]_i = \bigwedge^2 \eta_{i'} &= \pi^{-i'} \left(\bigwedge^2 (\Pi^{i'} r \underline{\mathcal{O}}^2) \right) \\ &= \pi^{-i-n} \bigwedge^2 (\Pi^{i+n} r \underline{\mathcal{O}}^2) \\ &= \pi^{-i} \bigwedge^2 (\Pi^{i+n} r g^{-1} \underline{\mathcal{O}}^2) \\ &= \pi^{-i} \bigwedge^2 (\Pi^{i} r_1 \underline{\mathcal{O}}^2).\end{aligned}$$

It is clear that the formula of the proposition defines a morphism of F into itself. To show that it is the one we want, we verify that $s = [M]$ and $g \in \mathrm{GL}_2(K)$, the following diagram is commutative:

$$\begin{array}{ccc} F_s(B) & \longrightarrow & F(B) \\ \downarrow g & & \downarrow g \\ F_{gs}(B) & \longrightarrow & F(B) \end{array}$$

(checking this is sufficient, since the union of the images of the F_s is dense and open). (See 8.2.2 for details.)

For $(\mathcal{L}, \alpha) \in F_s(B)$, we could obtain: $g \cdot (\mathcal{L}, \alpha) = (\mathcal{L}, \alpha \circ g^{-1})$. We can express the two cases (odd and even) via a similar formula: the image of $(\mathcal{L}, \alpha)$ in $F(B)$ is given by the following diagram:

$$\begin{array}{ccccc} \eta_e = \underline{M} & \xrightarrow{\Pi=\pi} & \eta_{e+1} = \underline{M} & \xrightarrow{\Pi=\mathrm{id}} & \eta_e = \underline{M} \\ \downarrow \alpha & & \downarrow \alpha & & \downarrow \alpha \\ T_e = \mathcal{L} & \xrightarrow{\Pi=\pi} & T_{e+1} = \mathcal{L} & \xrightarrow{\Pi=\mathrm{id}} & T_e = \mathcal{L} \end{array}$$

where $e = \log_q[\bigwedge^2 M : \bigwedge^2 \mathcal{O}^2]$ denotes the exponent of the (virtual) index of $\mathcal{O}^2$ in M (we do not assume here that M is normalised for $e \in \{0, 1\}$). The "rigidification" r is the composite of the morphism $K^2 \xrightarrow{\sim} \eta_e \otimes K$ associated with the inclusion $M \subset K^2$, with the morphism $\Pi^{-e} : \eta_e \otimes K \to \eta_0 \otimes K$.

The image of $(\mathcal{L}, \alpha \circ g^{-1})$ corresponds with the diagram:

$$\begin{array}{ccccc} \eta'_{e-n} = g\underline{M} & \xrightarrow{\Pi=\pi} & \eta'_{e-n+1} = g\underline{M} & \xrightarrow{\Pi=\mathrm{id}} & \eta'_{e-n} = g\underline{M} \\ \downarrow \alpha\circ g^{-1} & & \downarrow \alpha\circ g^{-1} & & \downarrow \alpha\circ g^{-1} \\ T'_{e-n} = \mathcal{L} & \xrightarrow{\Pi=\pi} & T'_{e-n+1} = \mathcal{L} & \xrightarrow{\Pi=\mathrm{id}} & T'_{e-n} = \mathcal{L} \end{array}$$

and with $r' = K^2 \xrightarrow{\sim} \eta'_0 \otimes K$ obtained by composing $K^2 \cong \eta'_{e-n}$ (associated with the inclusion $gM \hookrightarrow K^2$) with Π^{-e+n}.

We see, via the isomorphism $g : M \xrightarrow{\sim} gM$, the point (η', T', u', r') obtained from (η, T, u, r) is the one predicted in the proposition. □

Remark 8.2.2. *By Proposition 2.2.3, we should show that:*

1. *the action of g induces a morphism on $\widehat{\Omega}$,*
2. *the formal scheme $\widehat{\Omega}$ is reduced and separated.*

*The first assertion can be proved using an argument similar to the proof of the Yoneda lemma. Note that for any $A, B \in$ **Compl**, a homomorphism $\varphi : A \to B$ is uniquely determined by a compatible sequence of homomorphisms $\{\varphi_i\}_{i \geq 1}$, where each $\varphi_i : A \to B/\pi^i B$ is induced by φ.*

By Proposition 2.2.2, the second assertion is a direct consequence of the following lemmas.

Lemma 8.2.3. *We have $k[\zeta, \eta]/(\zeta\eta)$ is reduced.*

Proof. It is clear. □

Lemma 8.2.4. *The scheme $\widehat{\Omega}_k$ is separated.*

Proof. Since $\widehat{\Omega}_k$ is a union of increasing affine open subsets, the assertion follows immediately. □

References

[1] S. Bosch, *Lectures on Formal and Rigid Geometry*, Lecture Notes in Mathematics **2105**, Springer International Publishing, Cham, 2014. DOI: 10.1007/978-3-319-04417-0.

[2] J.-F. Boutot and H. Carayol, *Uniformisation p-adique des courbes de Shimura : les théorèmes de Čerednik et de Drinfeld*, Astérisque **196-197** (1991), 45–158. DOI: 10.24033/ast.73.

[3] S. Dasgupta and J. Teitelbaum, *The p-adic upper half plane*, in: p-adic Geometry, University Lecture Series **45**, Amer. Math. Soc., 2008, pp. 65–121.